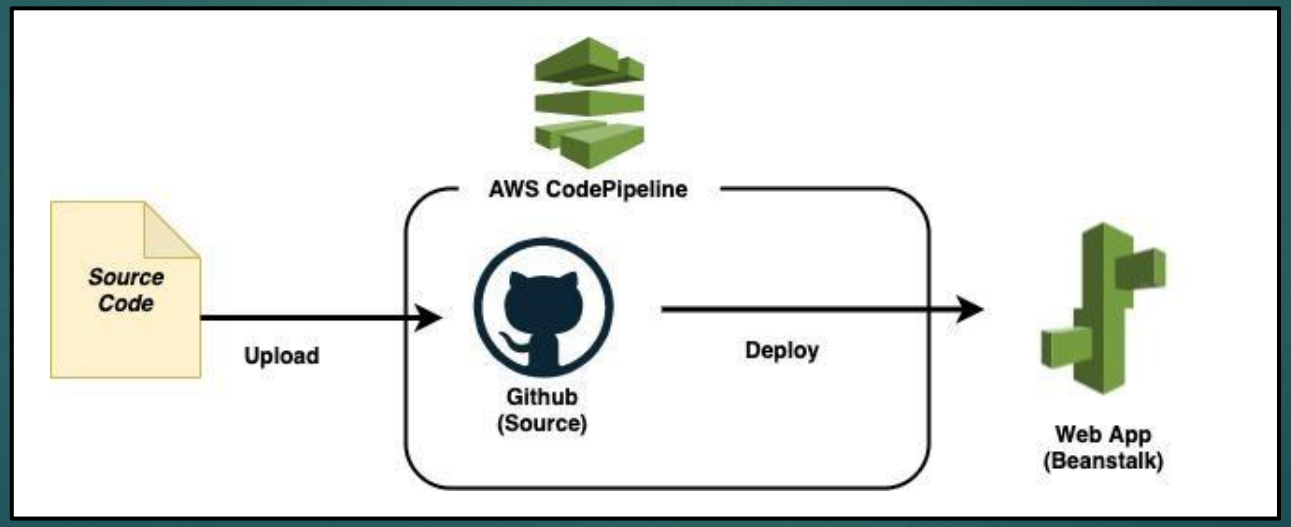
Create a DevOps Pipeline

SWEN 514: Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

Overview

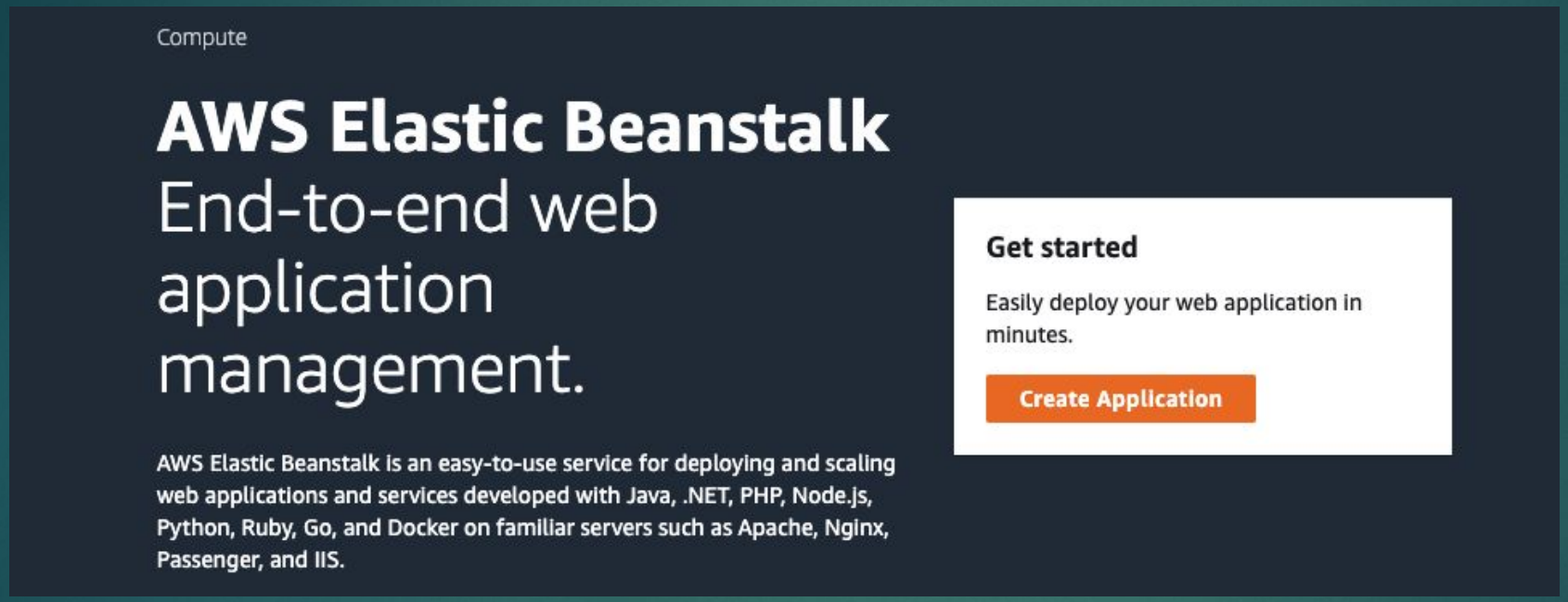
- In this activity, you create a Continuous Deployment pipeline that deploys a live sample web application
- This will use AWS CodePipeline to deploy code from your Github account to an Elastic Beanstalk NodeJS application
- Below is an illustration of the process



- There are 2 deliverables for this activity, which is worth 2 points

Create an app in Elastic Beanstalk

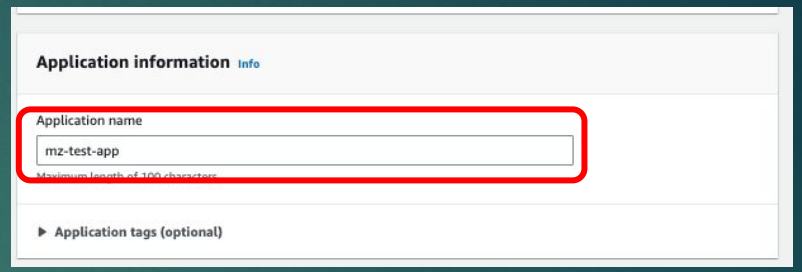
- Go to the AWS console and select "Elastic Beanstalk"

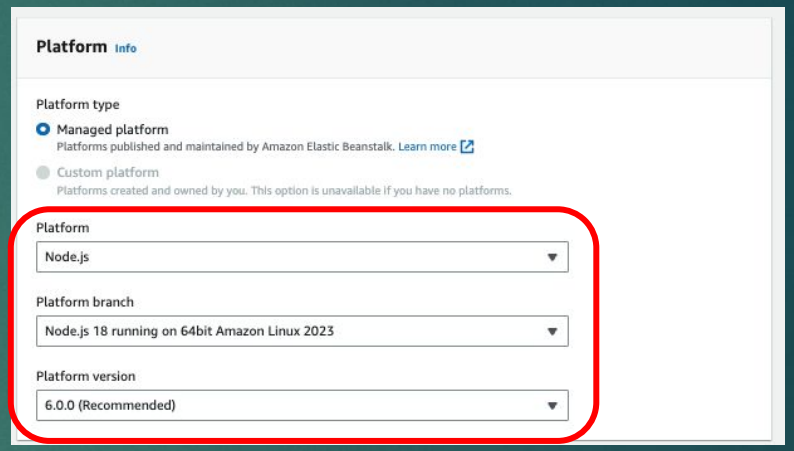


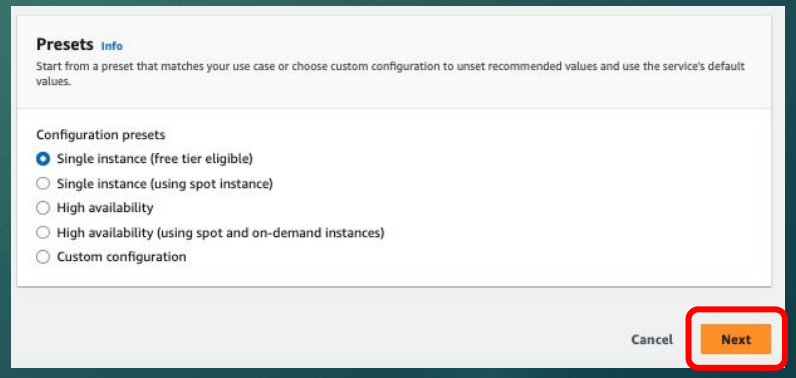
- Select the "Create Application" button

Create an app in Elastic Beanstalk

- Under “Application information” enter a name for “Application name”
- Under “Platform” select Node.js and keep all the defaults
- Under “Application information” put tags in “Application tags”
- Click “Next”

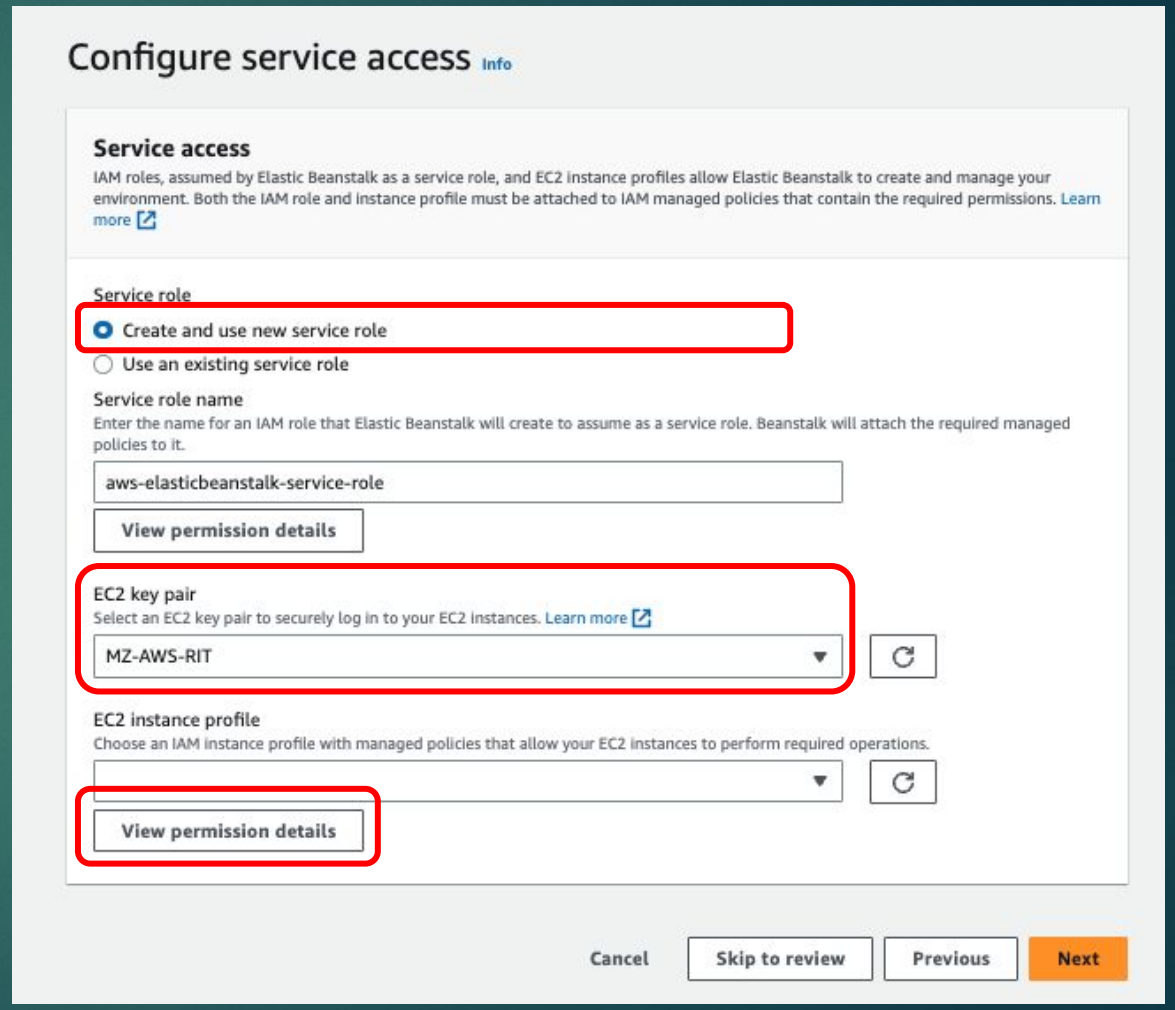






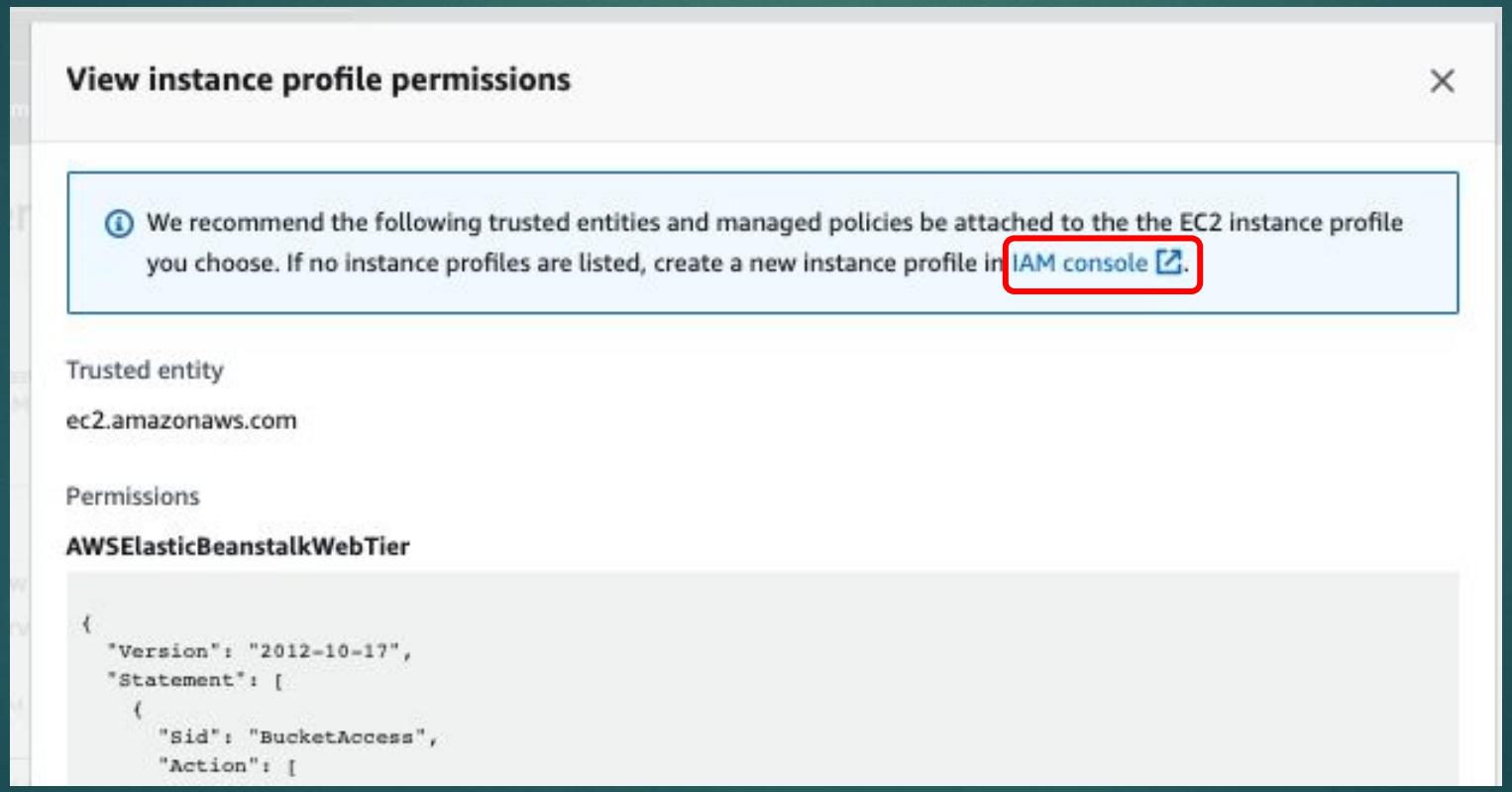
Create an app in Elastic Beanstalk

- Under "Service role", select "Create and use new service role"
  - Note: If you need to run this step again, the service role will be available so select "Use an existing service role" the next time since the role already is available
- Under "EC2 key pair", select your key pair
- Under EC2 instance profile, we need to create an IAM instance profile that will allow EC2 to communicate with other AWS services
  - Click the "View permission details" button



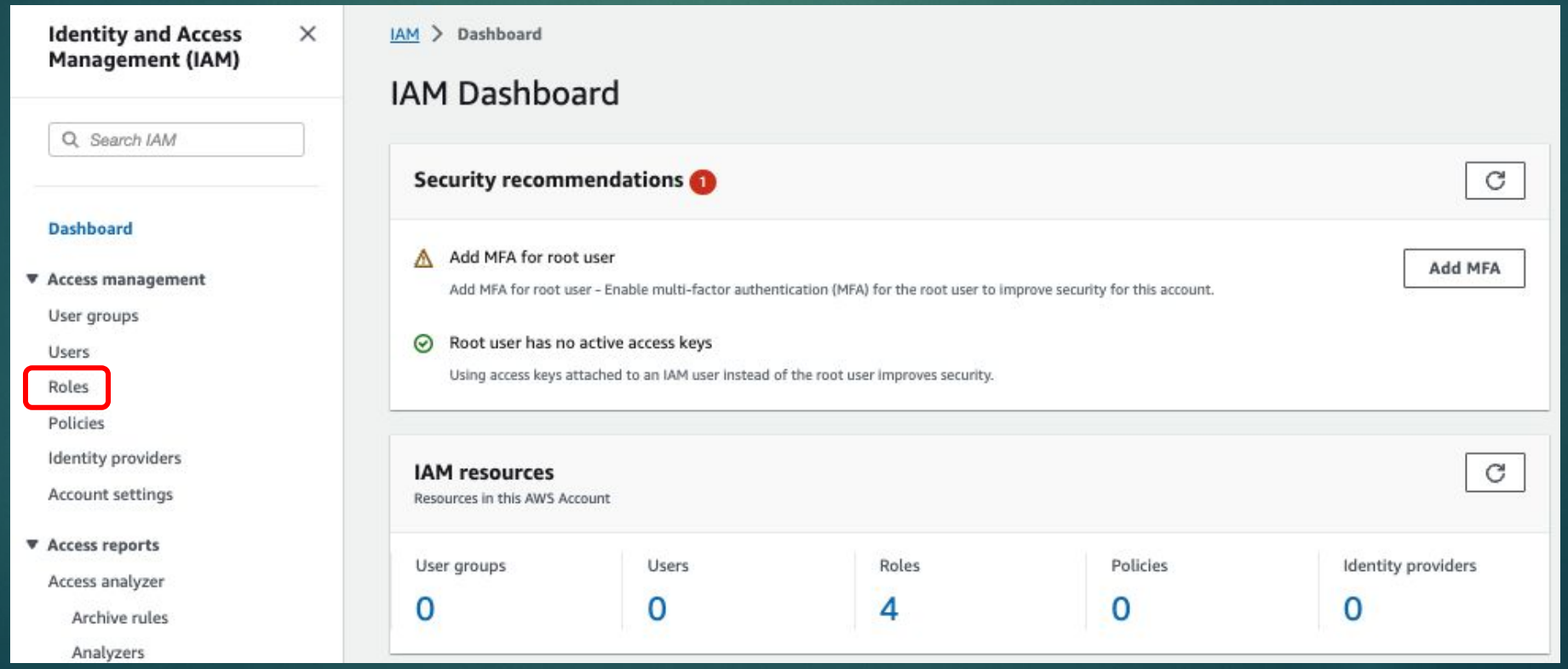
Create an app in Elastic Beanstalk

- We need to create a new instance profile to attach to the EC2 instance
- To do this, click the "IAM console"



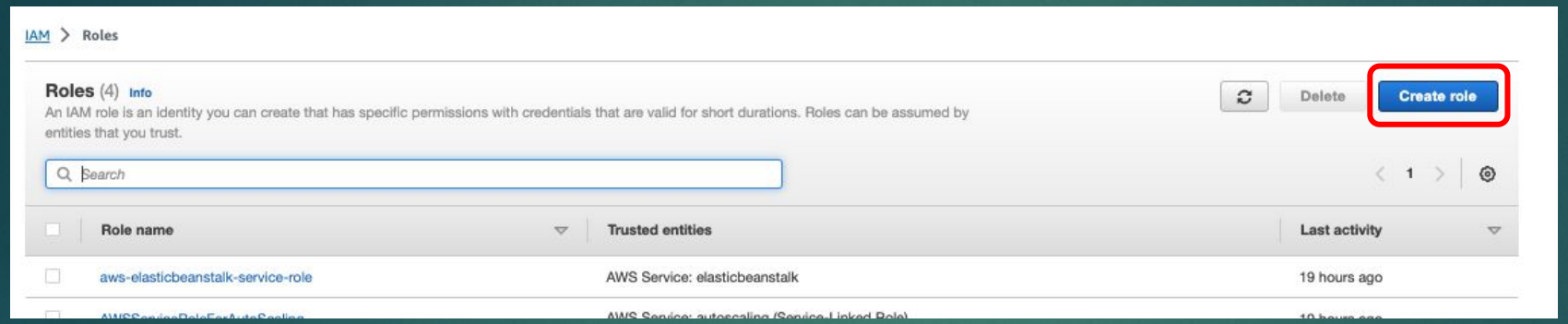
IAM Dashboard

- On the IAM Dashboard, we are going to create a new instance profile (role)
- Click the "Roles" on the left

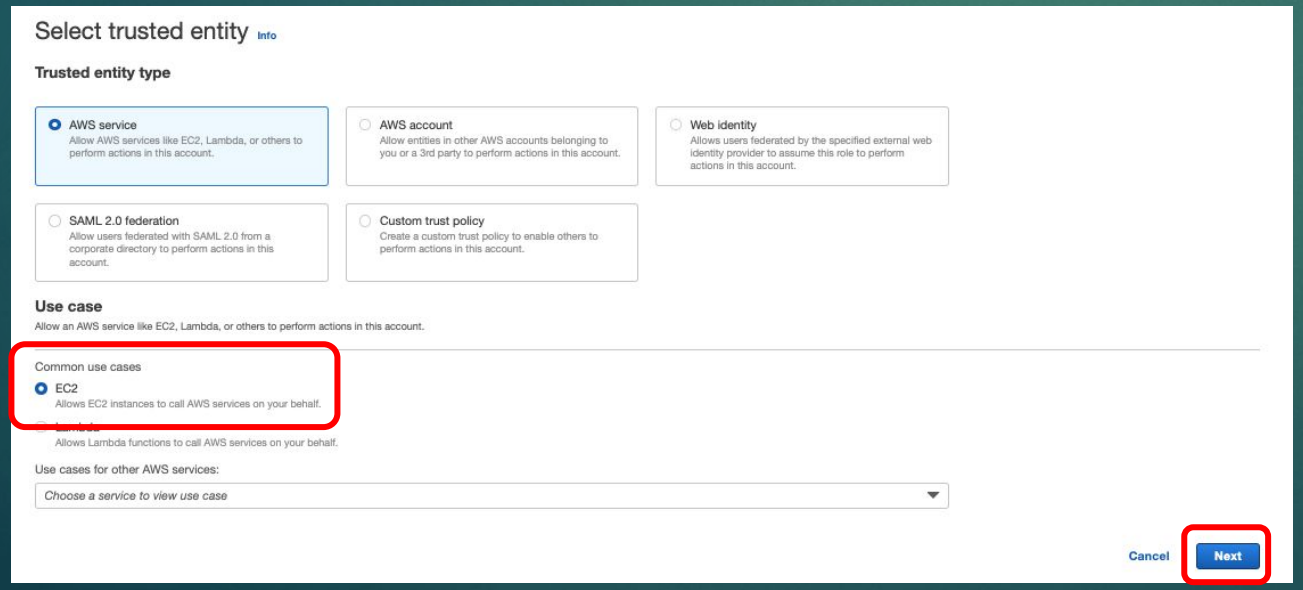


Create IAM Role

- Select "Create role"

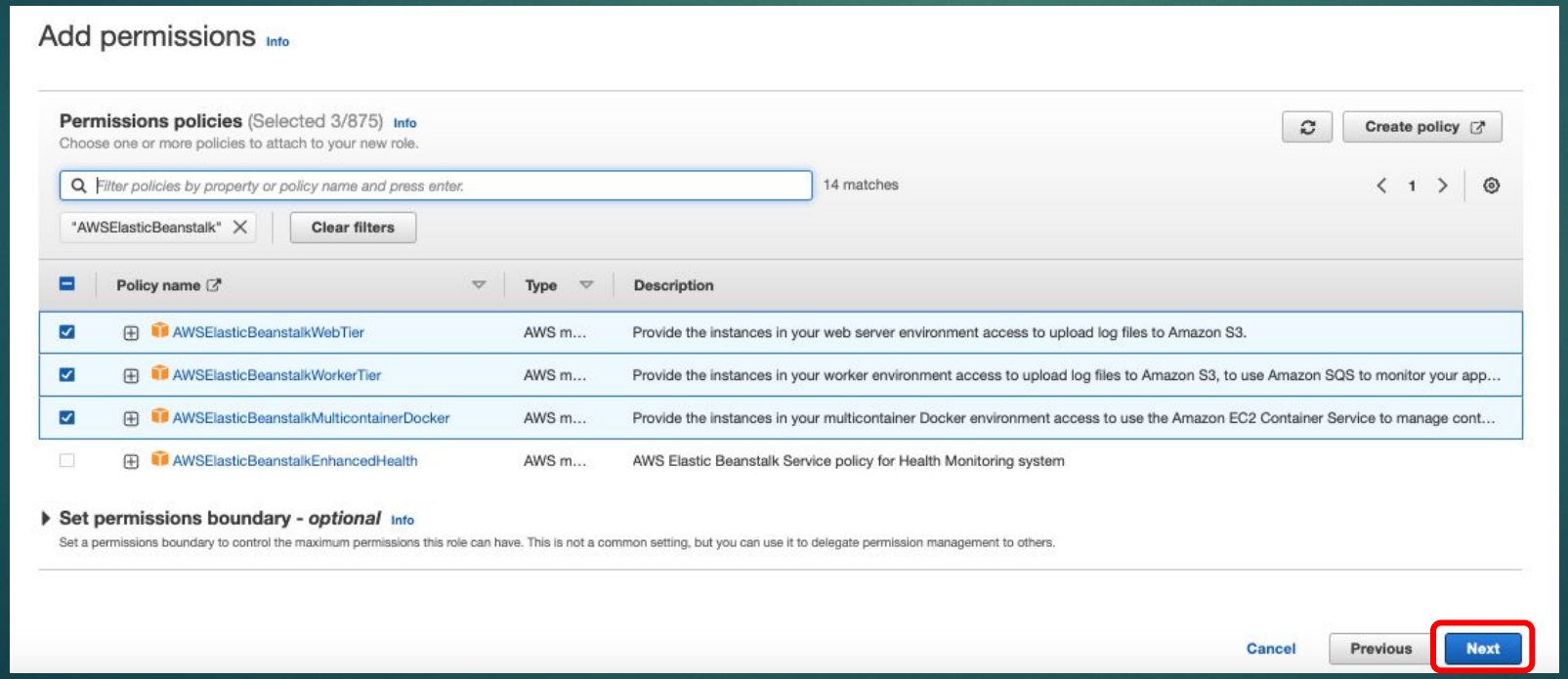


- On the next screen, select "EC2" under "Use case" and click "Next"



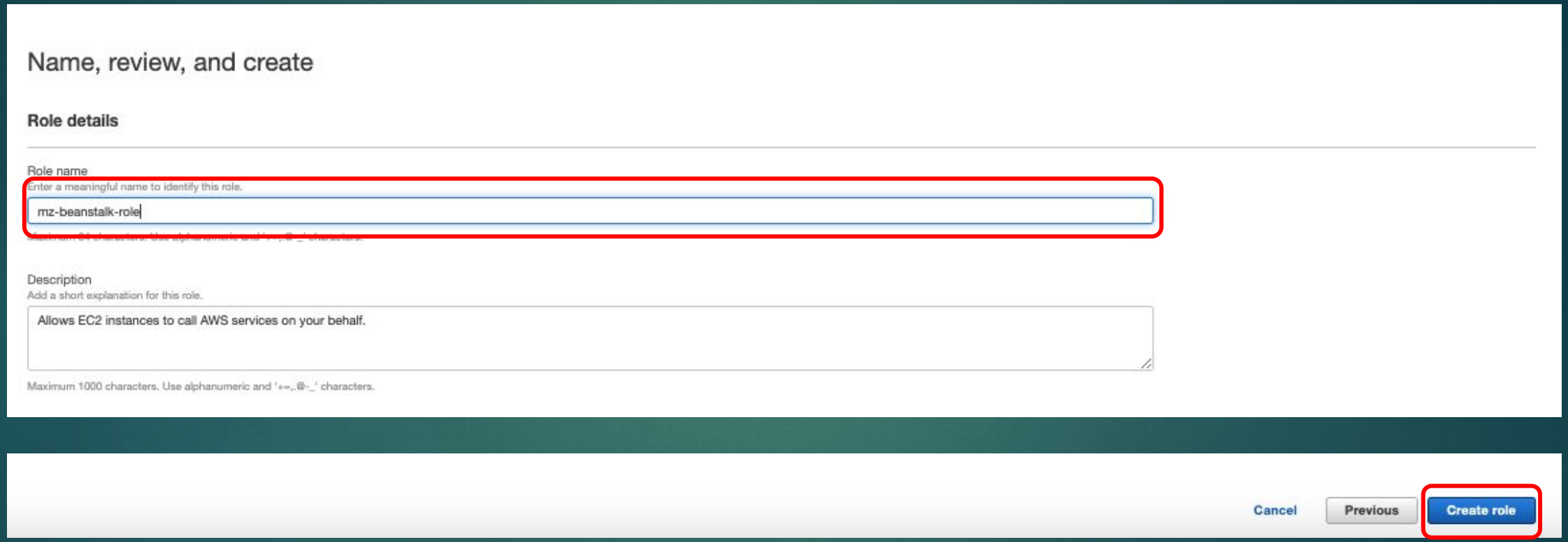
Create IAM Role

- In the search box, enter "AWSElasticBeanstalk" and hit Enter
- A subset of permissions is returned
- Select the top 3 permissions (see below)
- Click "Next"



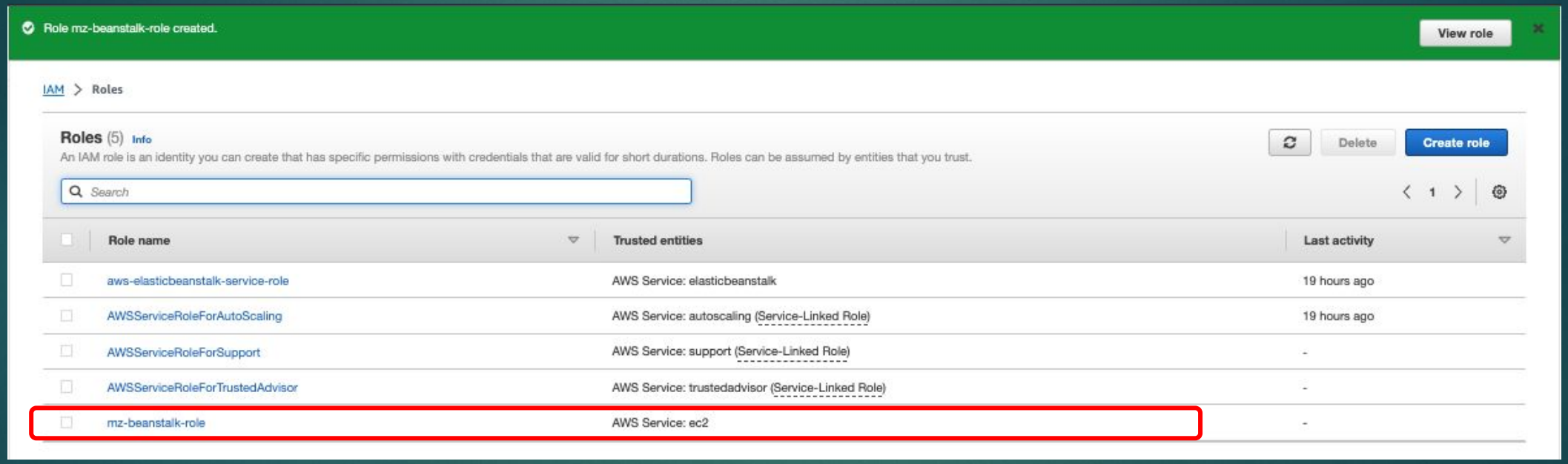
Create IAM Role

- Give your role a name and click "Create role"



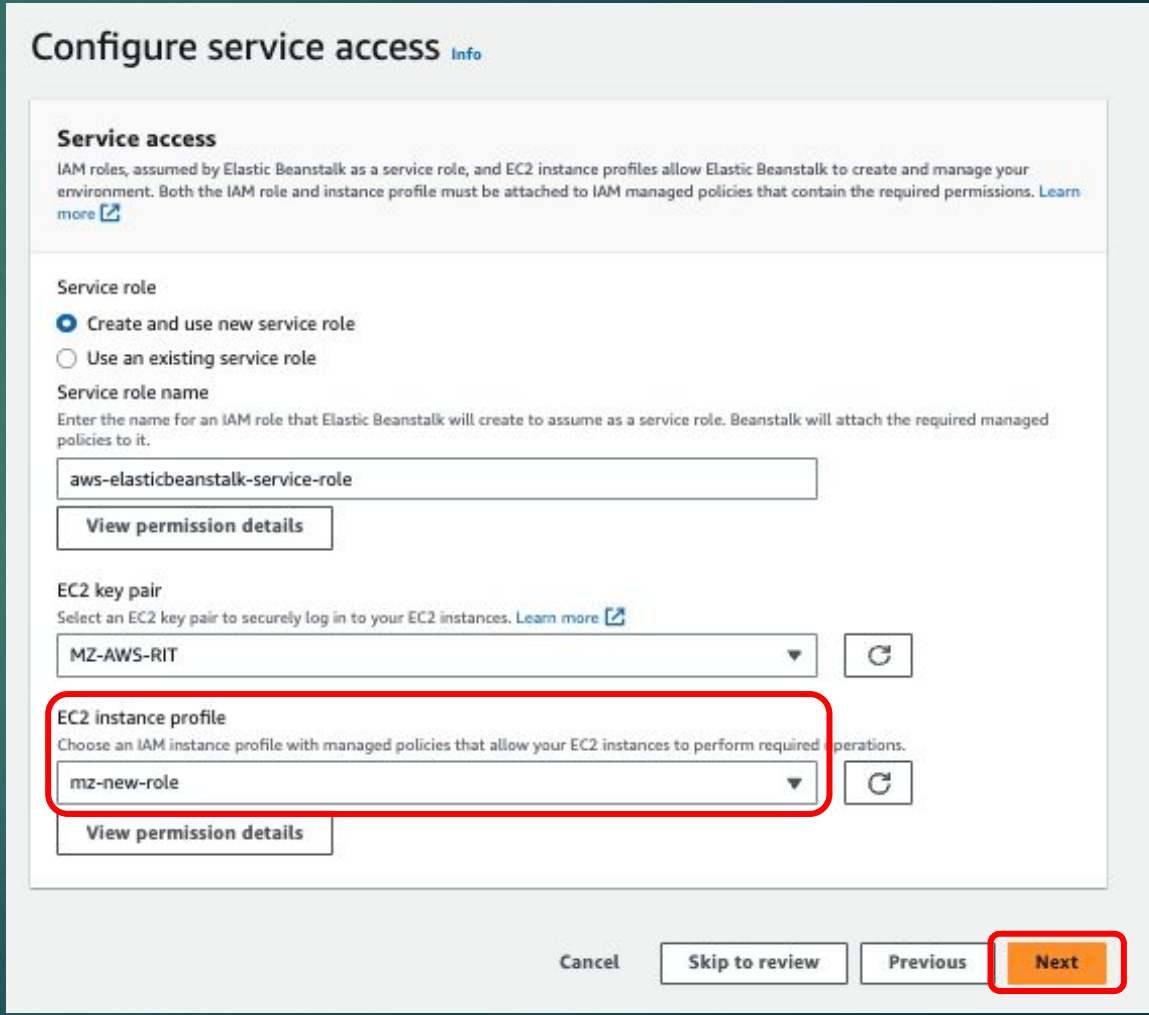
Create IAM Role

- Verify your role is created and go back the the tab where you were configuring your Beanstalk app



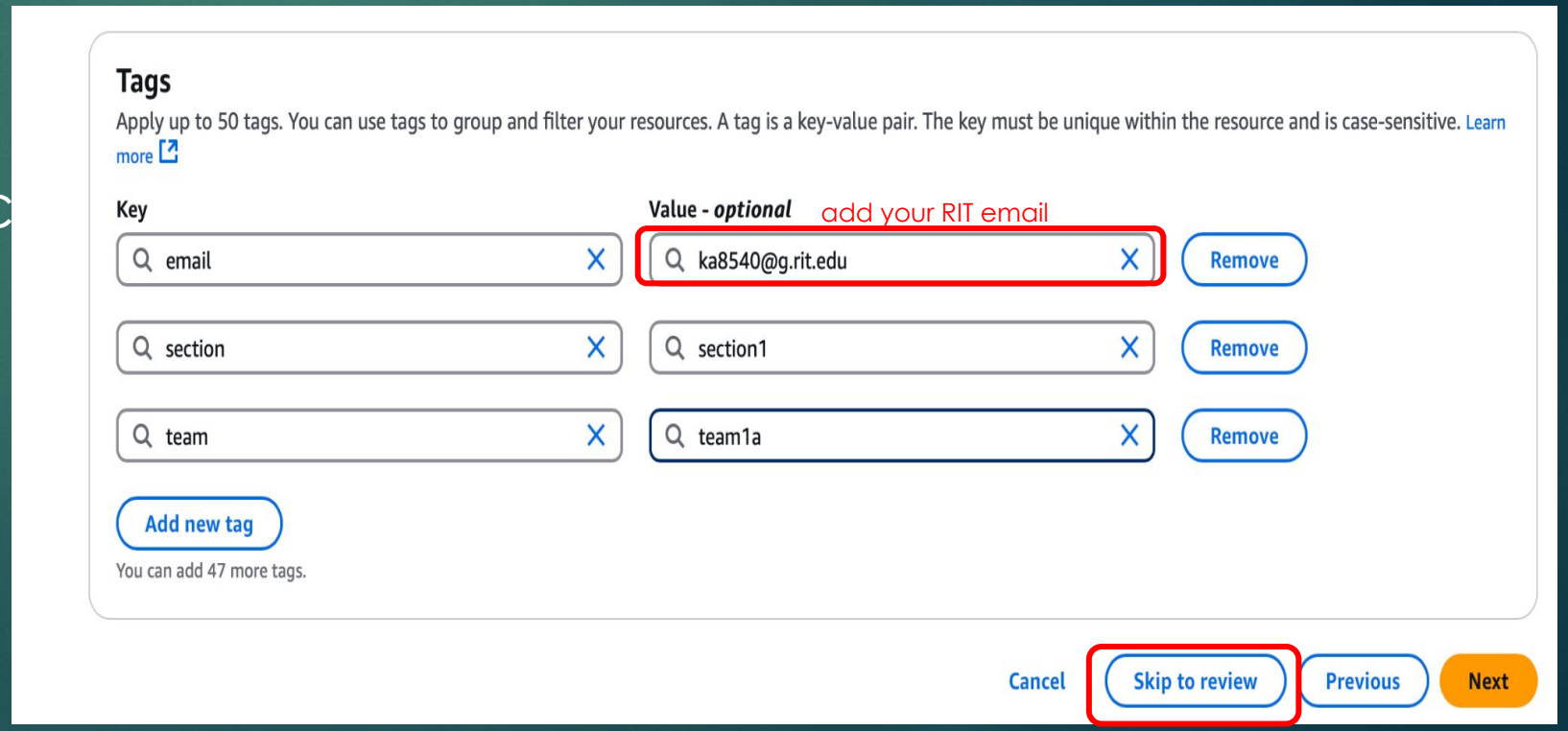
Create an app in Elastic Beanstalk

- Your new instance profile should now be available under "EC2 instance profile"
- Select it and click "Next"



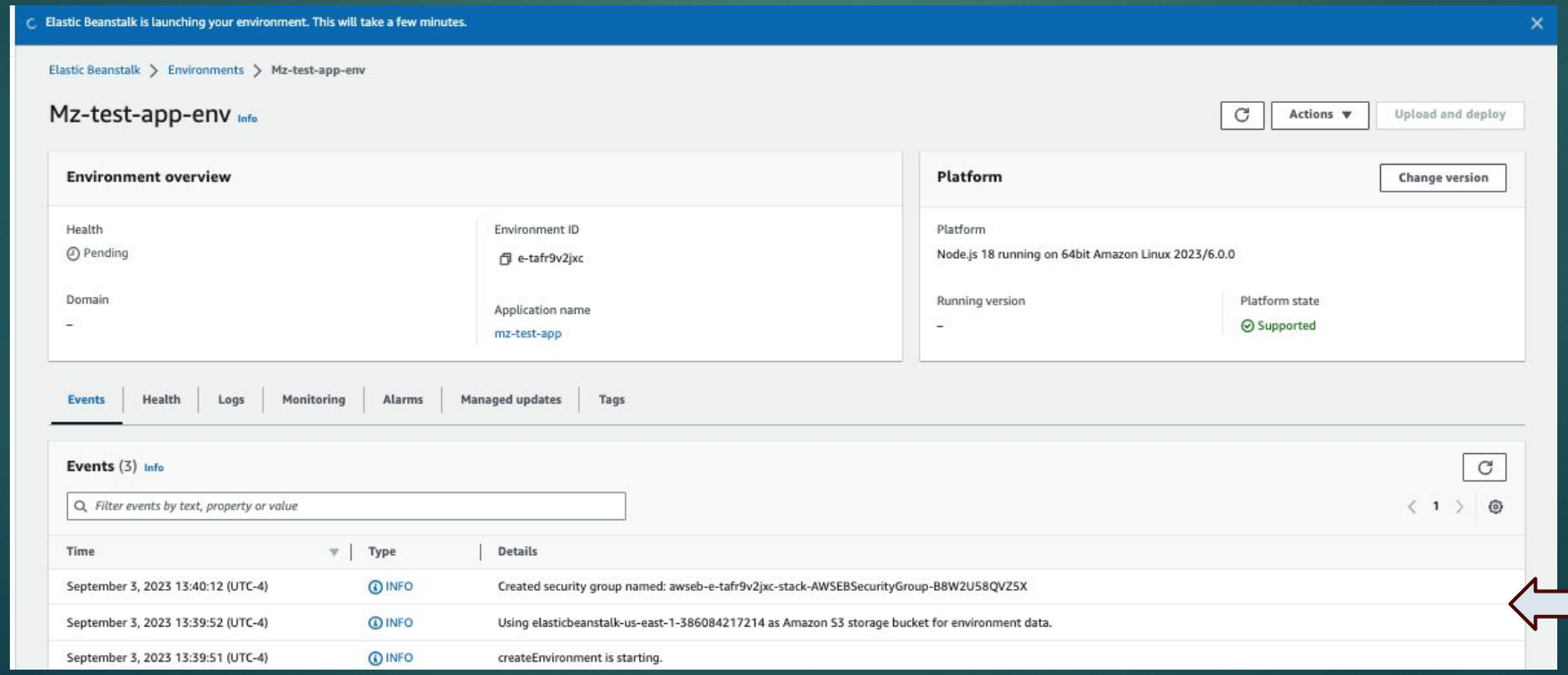
Create an app in Elastic Beanstalk

- At the bottom, select "**Add Tags**"
- add the specified tags (email, section, team)
- and click "Skip to review"
- On the Review screen, clic "Submit" to create your new application



Create an app in Elastic Beanstalk

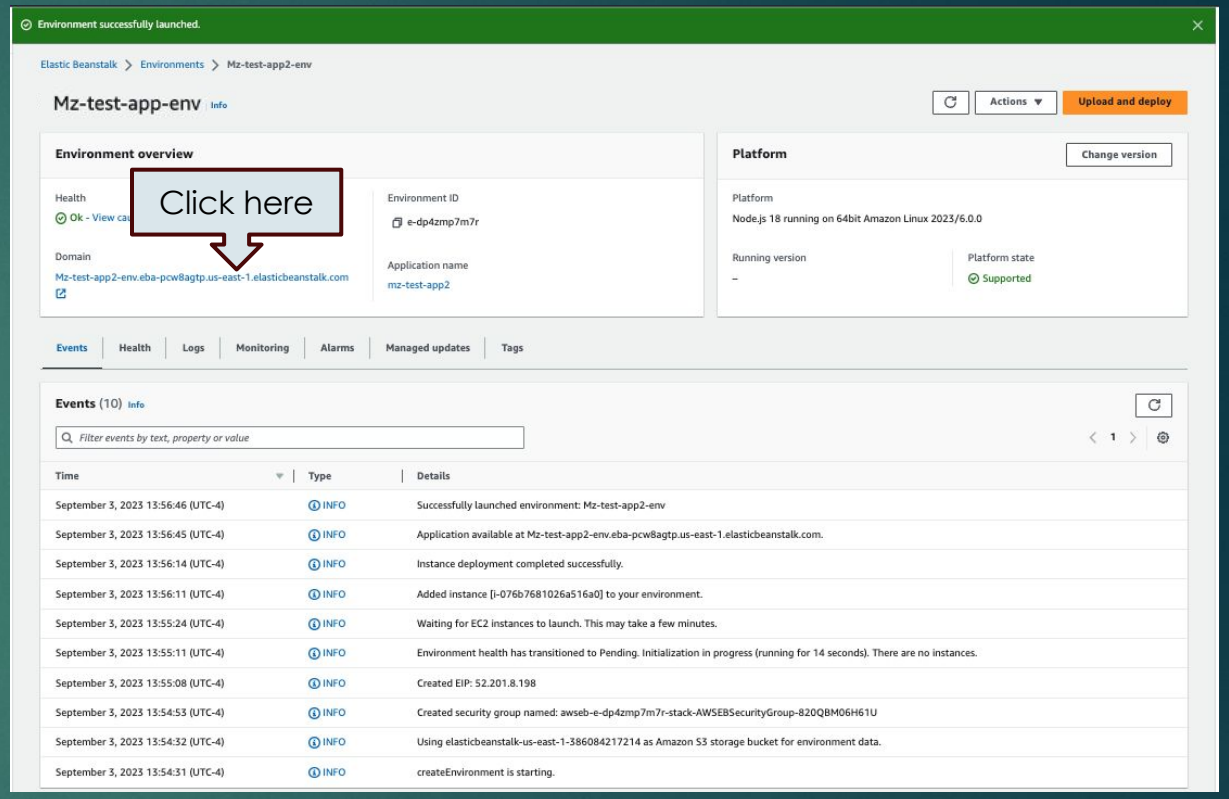
- Your app is now being created and will take about 3-5 minutes
- Note the steps in "Events" are from a CloudFormation script that's creating the application
- Feel free to check CloudFormation as it will show several more steps



CloudFormation steps

Create an app in Elastic Beanstalk

- When completed, you should see a success message
- Click the link under "Domain"



- <u>Note:</u> If Beanstalk is taking longer than 5 minutes, do the following:
  - Go to CloudFormation and delete the Stack
  - Start over on slide #4

Create an app in Elastic Beanstalk

- The sample application has been deployed and is running

- Next, we will connect a source code repository (Github) to the application
- When the repository is modified, this will trigger the application to redeploy

Download source files

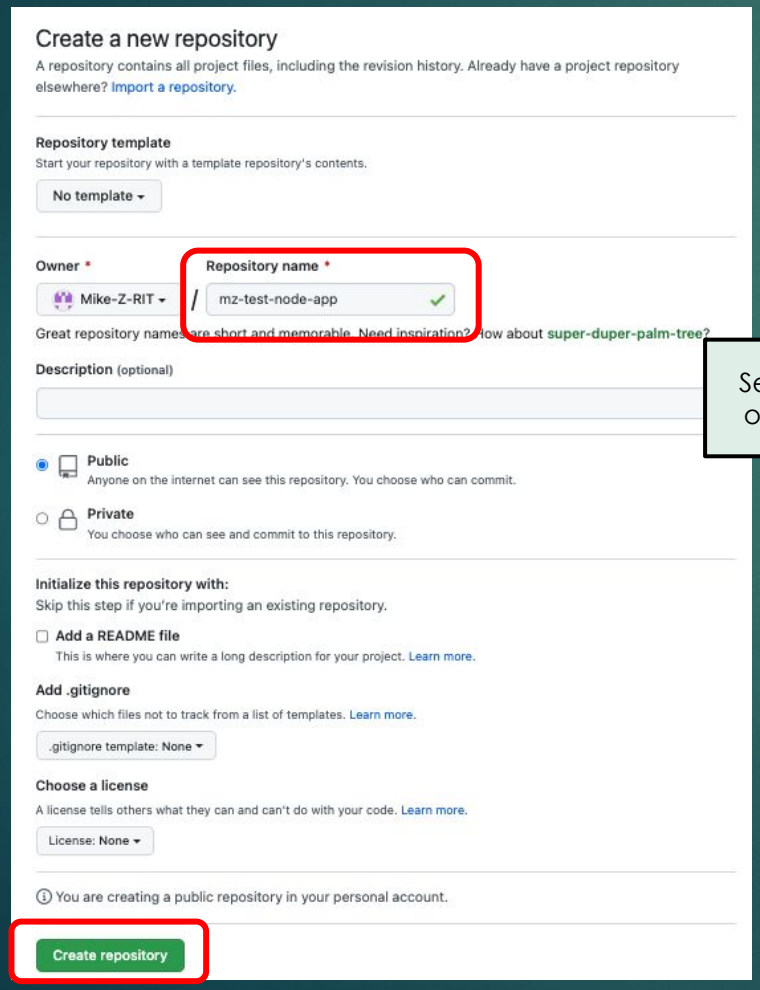
- The source for the sample app (nodejs.zip) can be found under Assignments > Activity #14 - Create a DevOps Pipeline
- Download this file to your PC and unzip

```
mikez@Mikes-MacBook-Pro-3 node-test2 % unzip nodejs.zip
Archive:  nodejs.zip
  inflating: EBSampleApp-Nodejs.iml
  inflating: cron.yaml
  inflating: index.html
  inflating: package.json
   creating: .ebextensions/
  inflating: .ebextensions/logging.config
  inflating: app.js
mikez@Mikes-MacBook-Pro-3 node-test2 %
```

- Next, go to Github to create a new repo

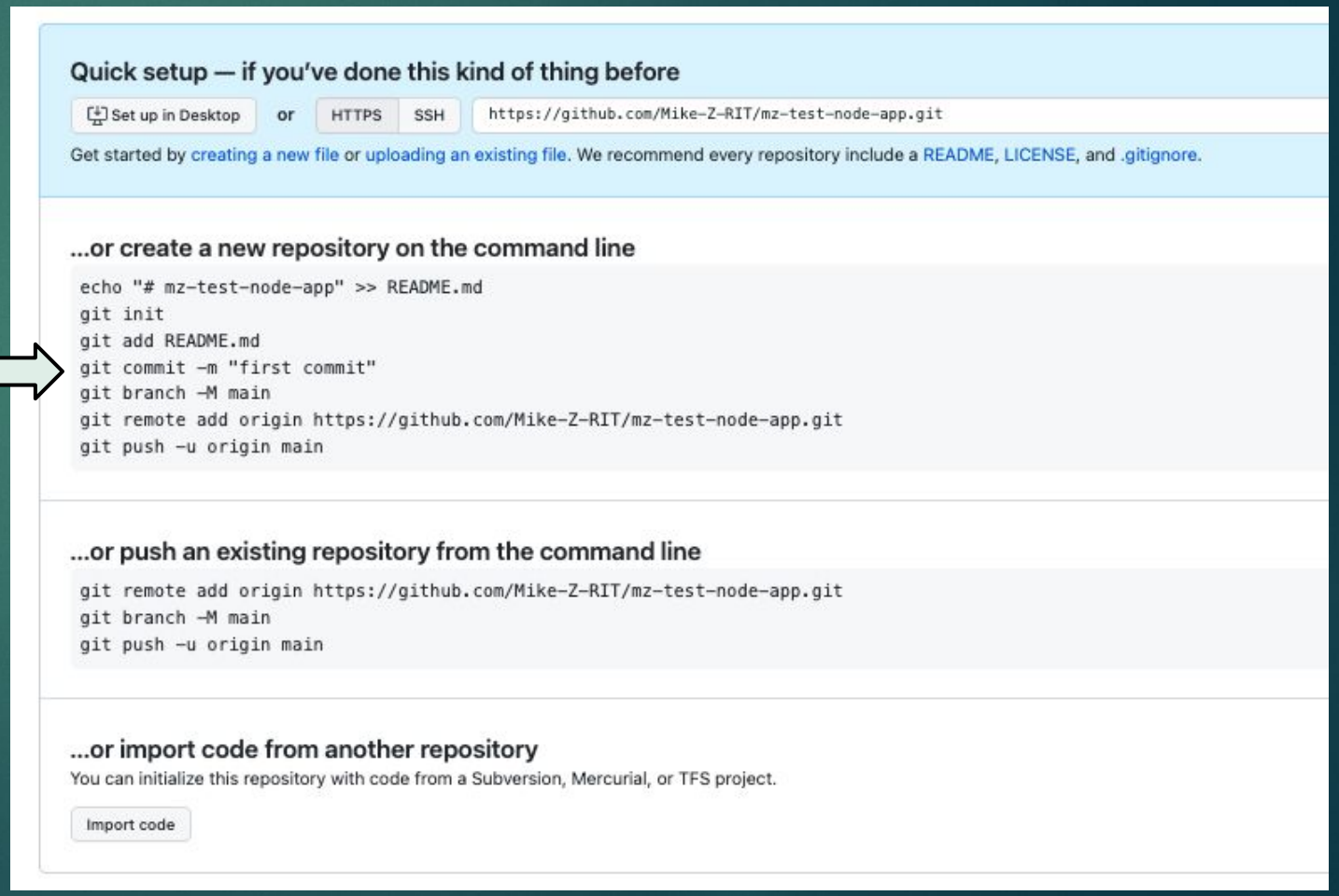
Create a Repo

- Create a new repo in Github and click "Create repository"



See example on next slide

- Follow the instructions to import the sample app to your repo

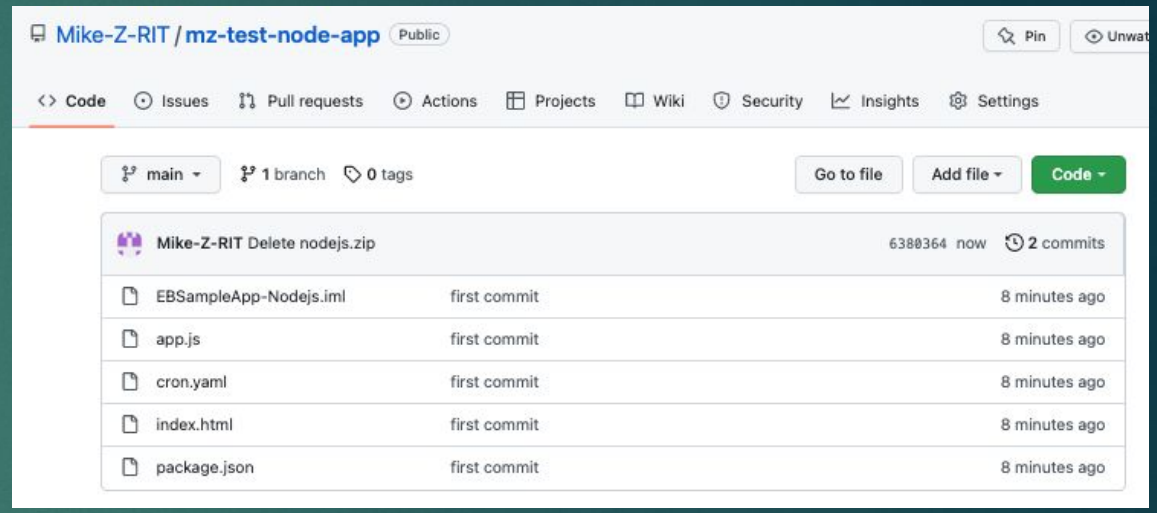


Create a Repo (example)

- Here is an example of pushing the files to Github following the steps from the previous slide

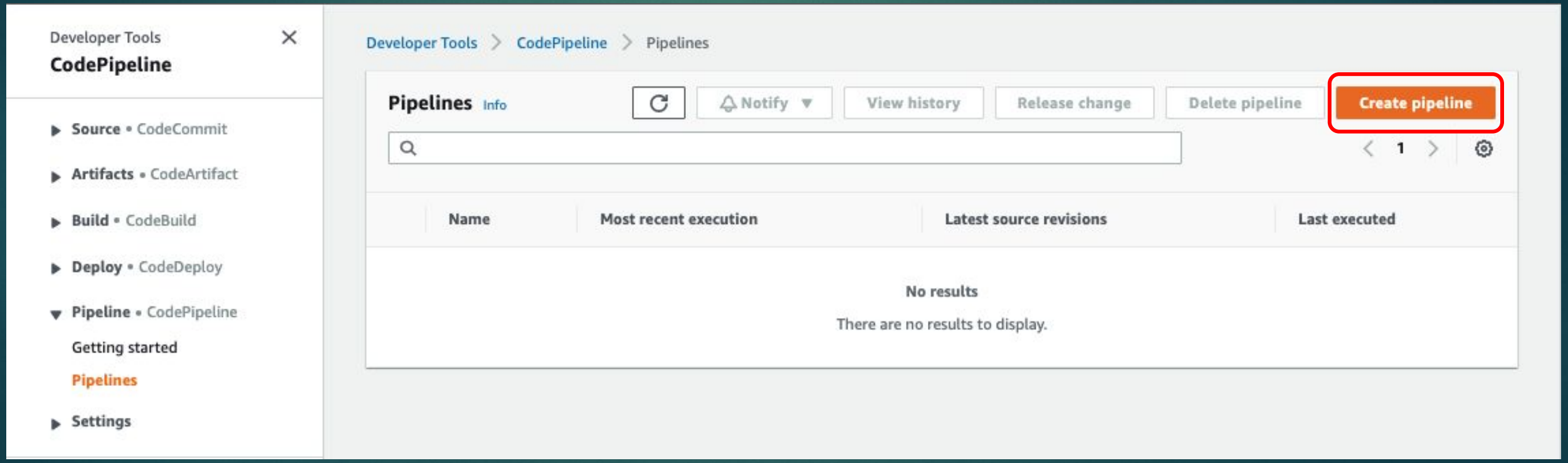
```
mikez@Mikes-MacBook-Pro-3 node-test2 % git init
Initialized empty Git repository in /Users/mikez/RIT/node-test2/.git/
mikez@Mikes-MacBook-Pro-3 node-test2 % git add *
mikez@Mikes-MacBook-Pro-3 node-test2 % git commit -m "first commit"
[master (root-commit) 440d2fa] first commit
 6 files changed, 163 insertions(+)
 create mode 100644 EBSampleApp-Nodejs.iml
 create mode 100644 app.js
 create mode 100644 cron.yaml
 create mode 100644 index.html
 create mode 100644 nodejs.zip
 create mode 100644 package.json
mikez@Mikes-MacBook-Pro-3 node-test2 % git branch -M main
mikez@Mikes-MacBook-Pro-3 node-test2 % git remote add origin https://github.com/Mike-Z-RIT/mz-test-node-app.git
mikez@Mikes-MacBook-Pro-3 node-test2 % git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads.
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 4.65 KiB | 1.55 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0)
To https://github.com/Mike-Z-RIT/mz-test-node-app.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
mikez@Mikes-MacBook-Pro-3 node-test2 %
```

- When complete, verify your repo has the files below



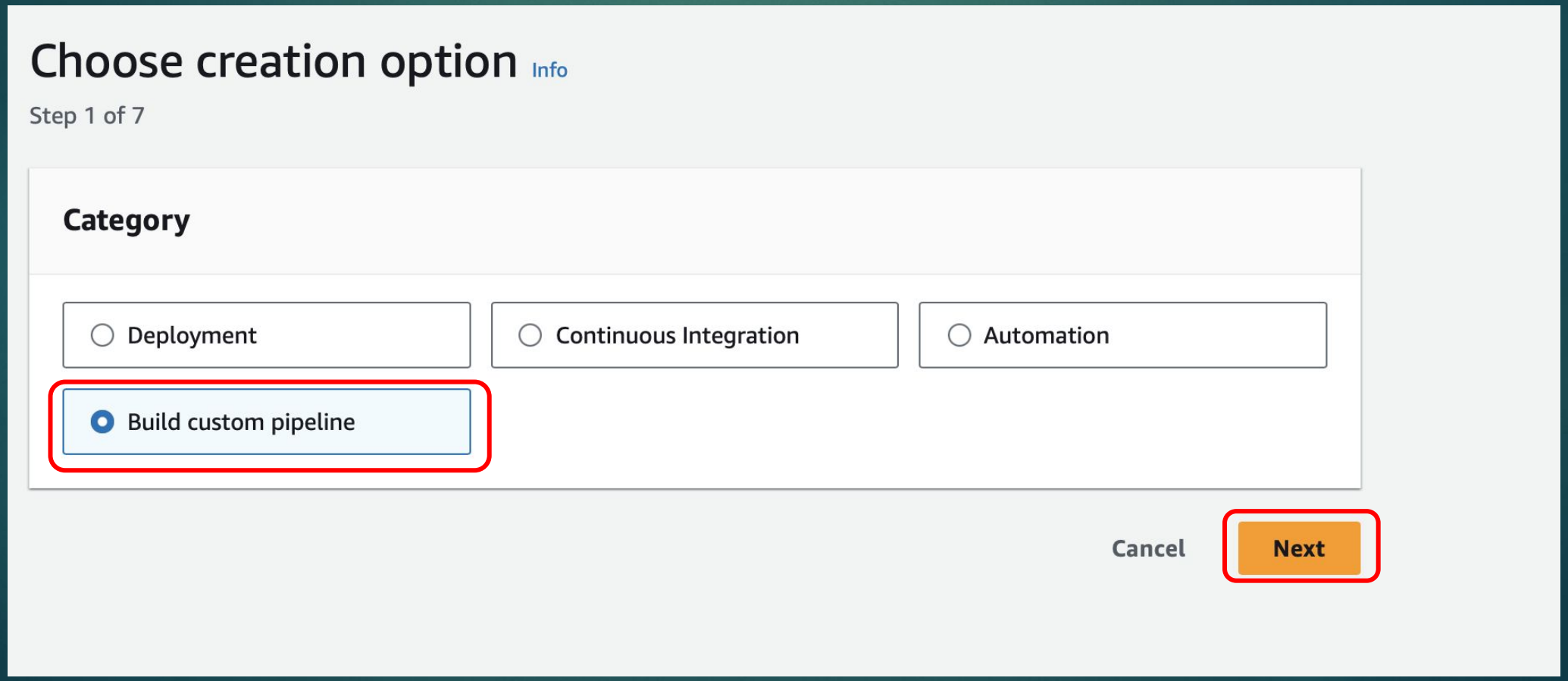
Create your Pipeline

- Go to the AWS console and select “CodePipeline”
- Click the “Create pipeline” button



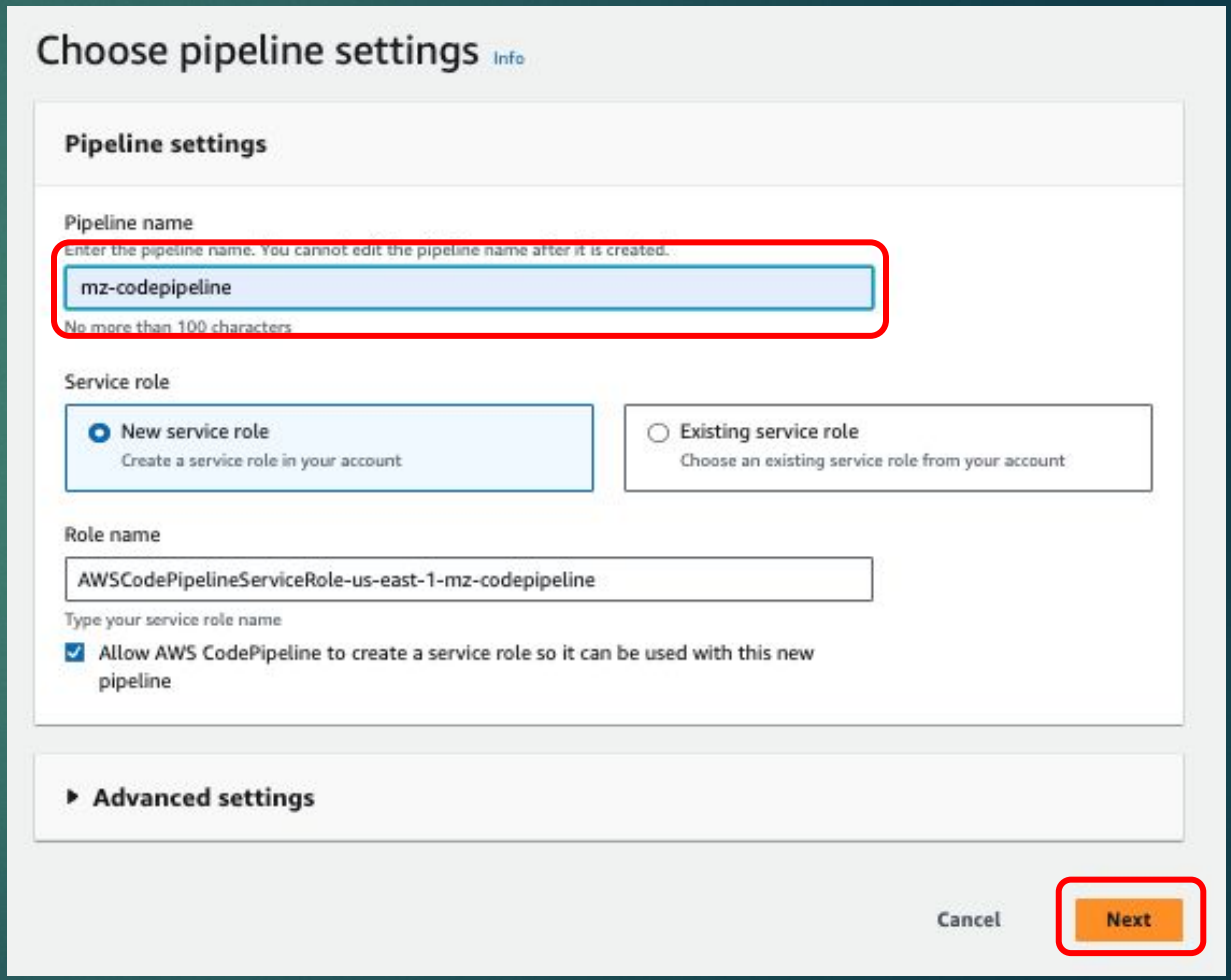
Create your Pipeline

- Under Choose creation option select “Build Custom Pipeline”
- Click “Next”



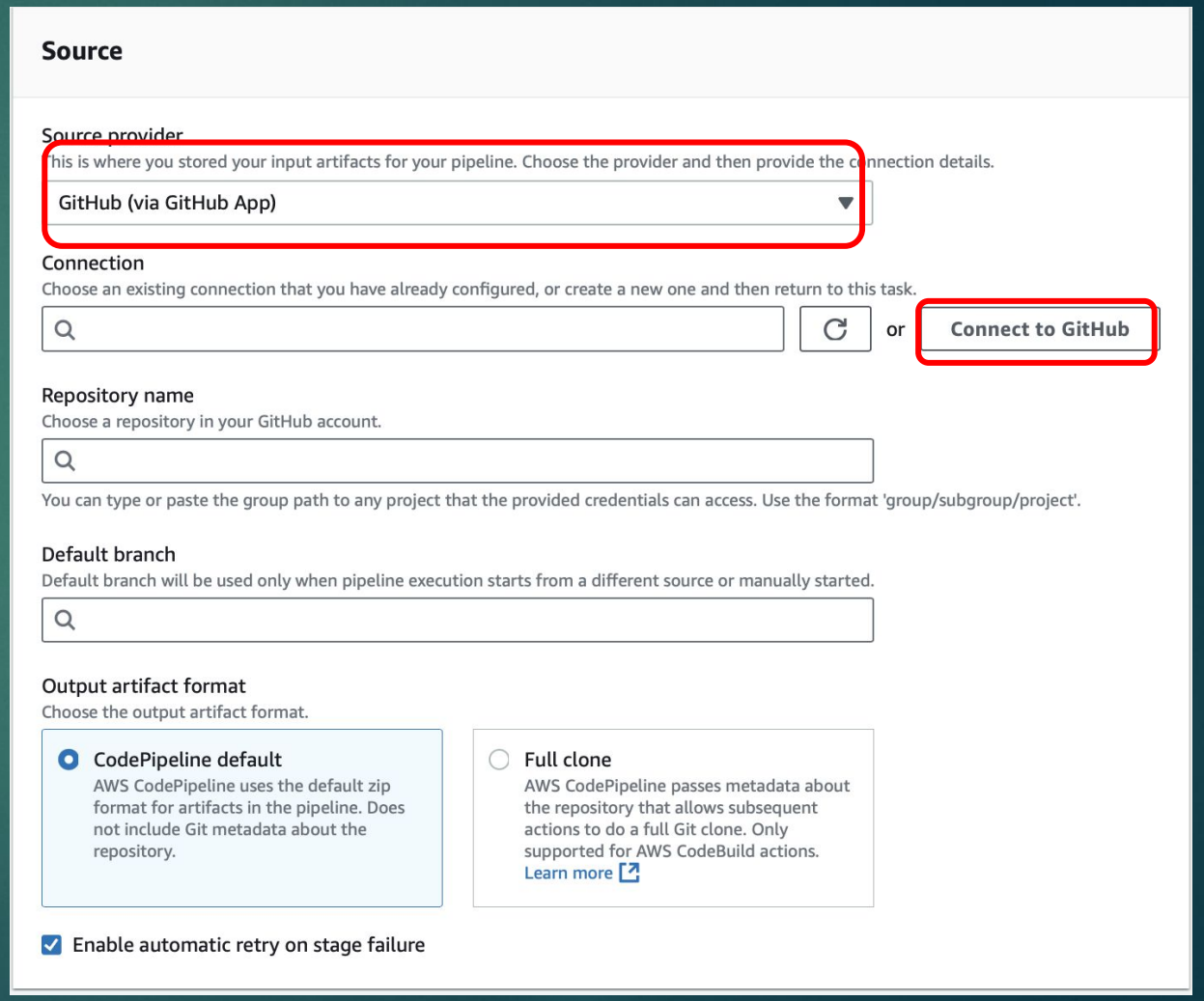
Create your Pipeline

- Enter name for "Pipeline name"
- Click "Next"



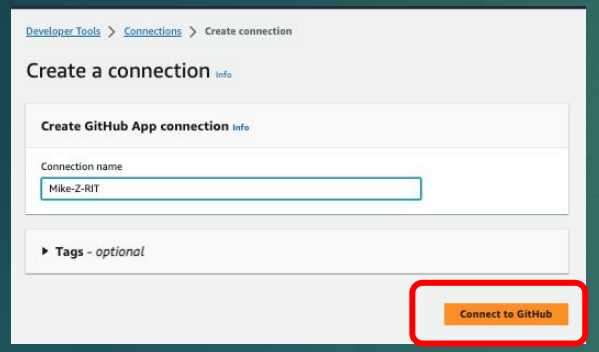
Pipeline Source

- Under "Source Provider", Select "Github (via Github APP)"
- Click the "Connect to Github" to link your AWS account to your Github account
- This is a multi-step process but only needs to be done once

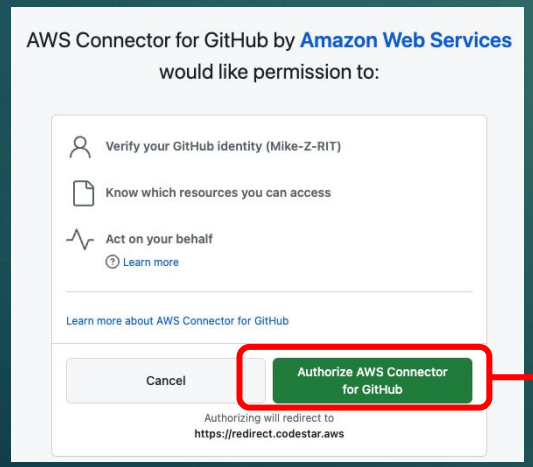


Connect Pipeline to Github

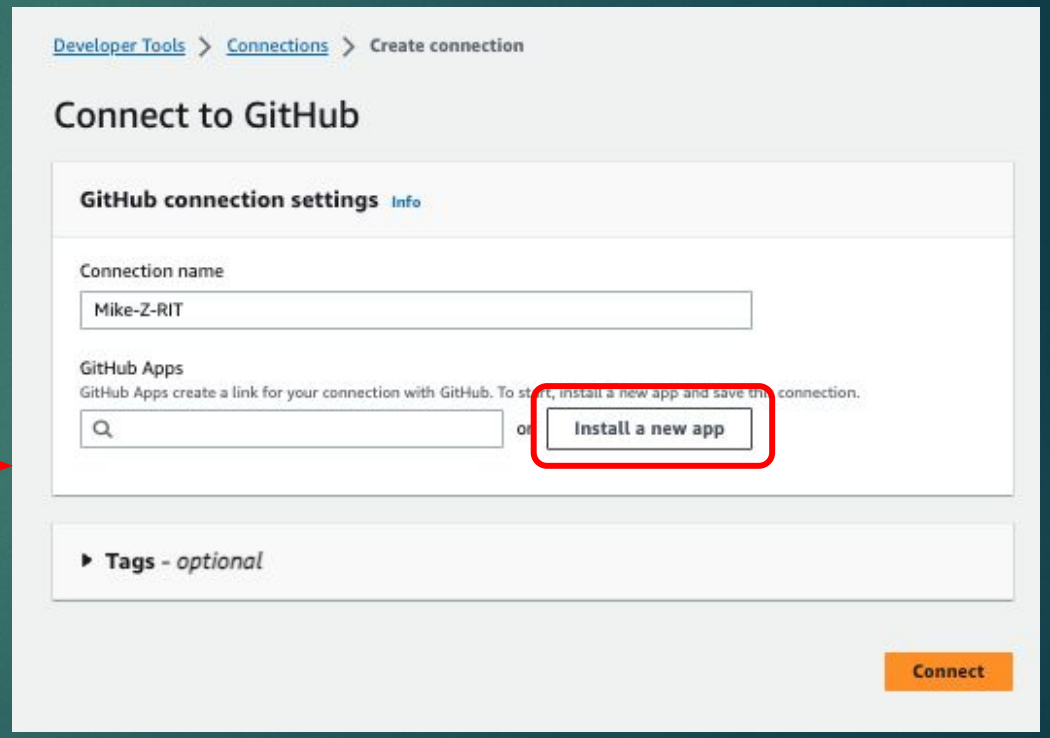
#1 - To connect to Github, provide your Github ID and click "Connect to Github"



#2 - Click the "Authorize AWS Connector for Github"

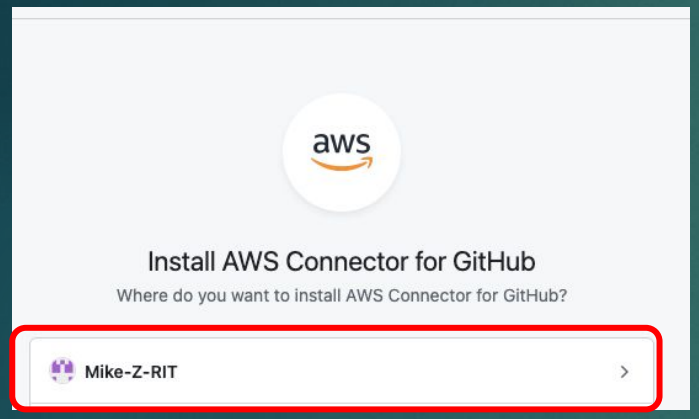


#3 - Click "Install a new app"

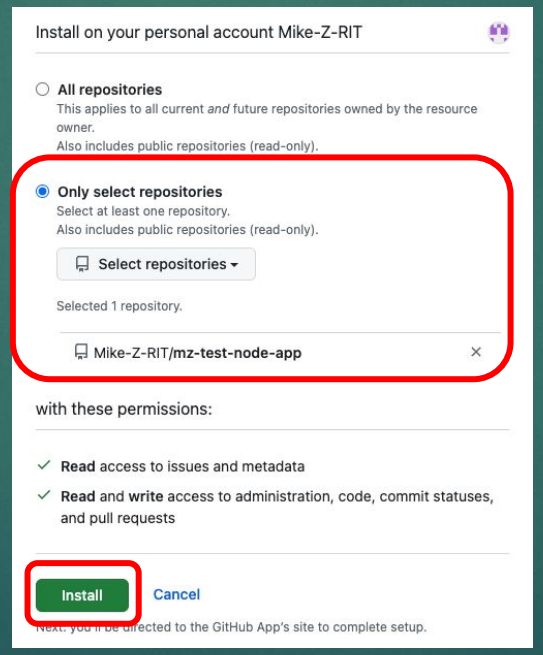


Connect Pipeline to Github

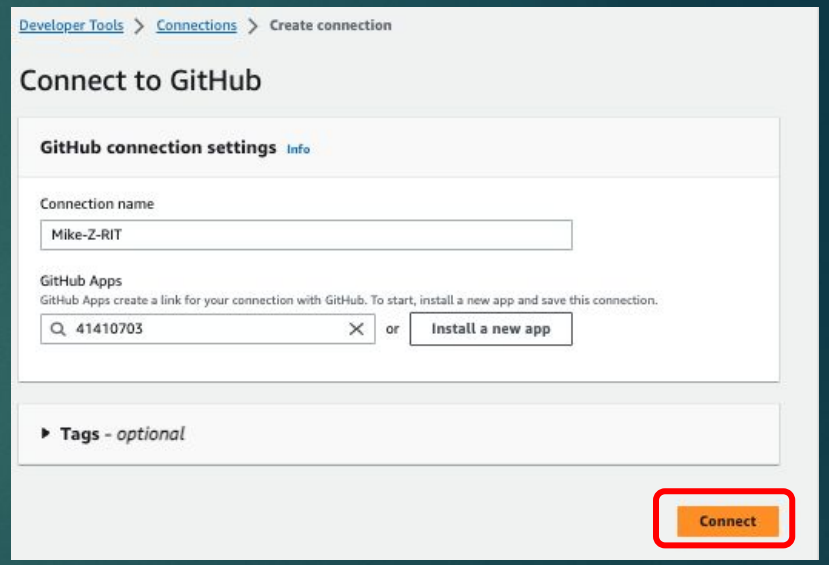
#1- Select your Github account



#2 - Select "Only select repositories" and select the repo you just created. Click "Install"

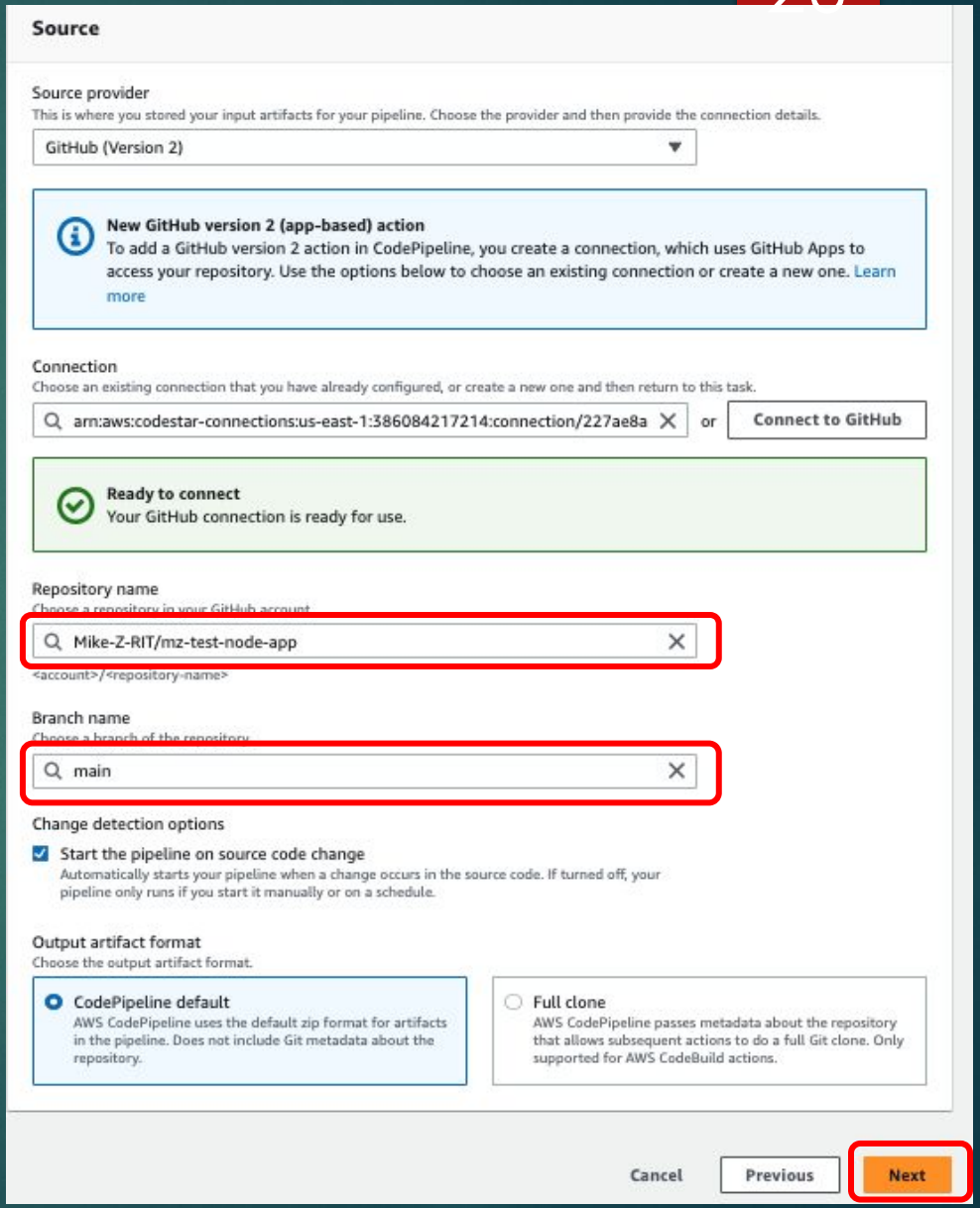


#3 - Click "Connect" to connect AWS to your Github account



Pipeline Source

- Pipeline is now connected to Github
- Select the repo you previously created
- Select the "main" branch to trigger a deploy
- Your screen should look similar to the right
- Keep Trigger Type as "**No Filter**"
- Click "Next"



Pipeline Source

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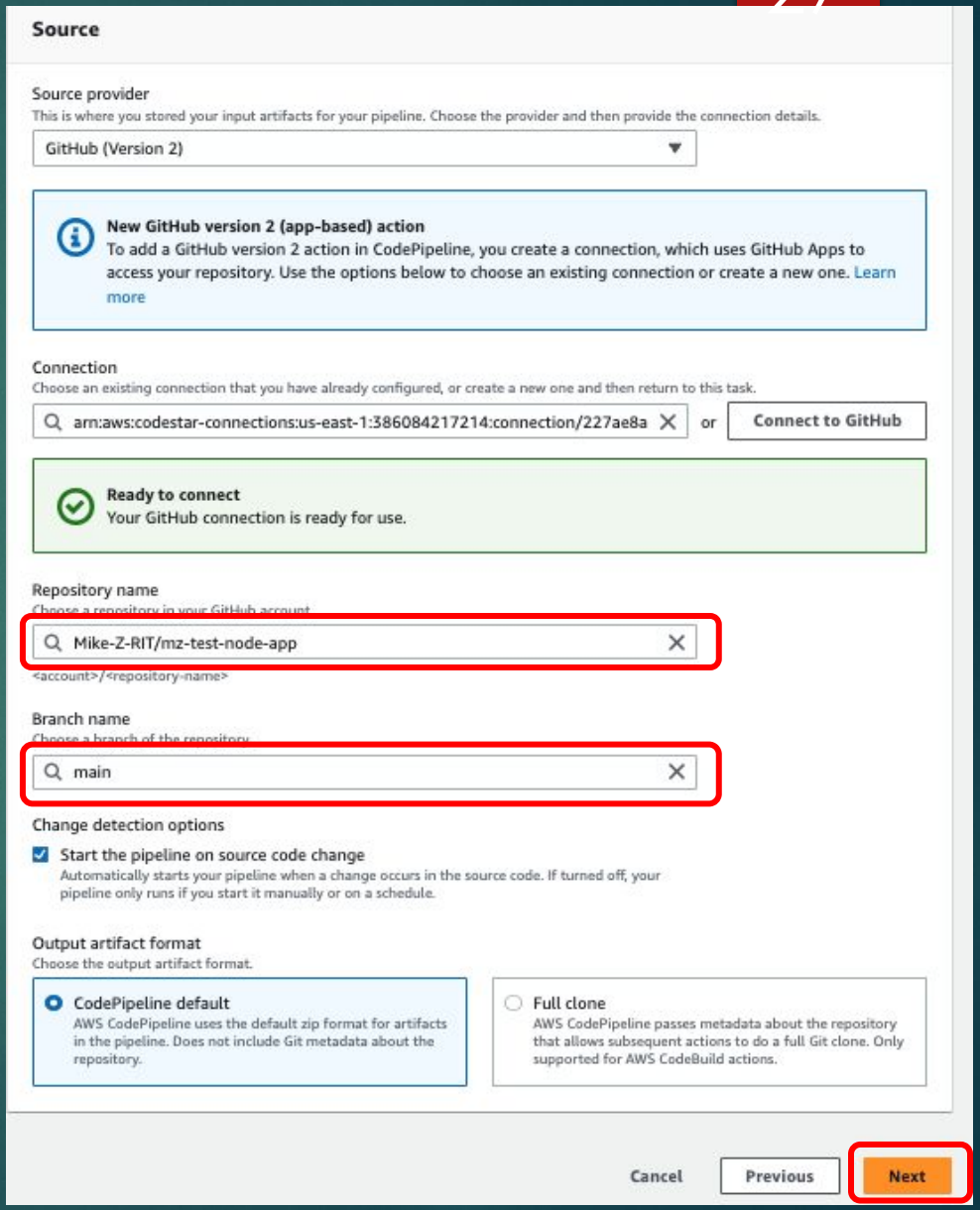
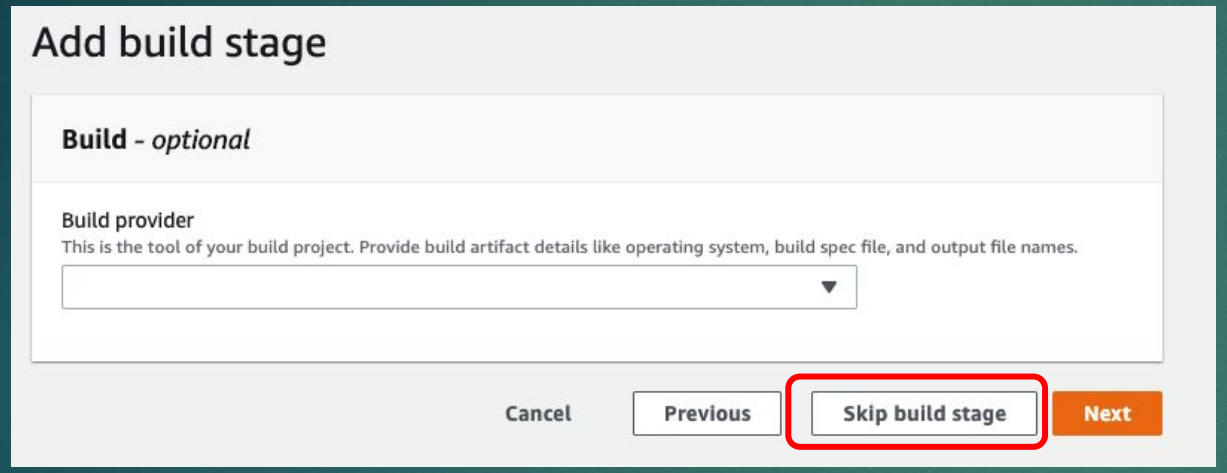
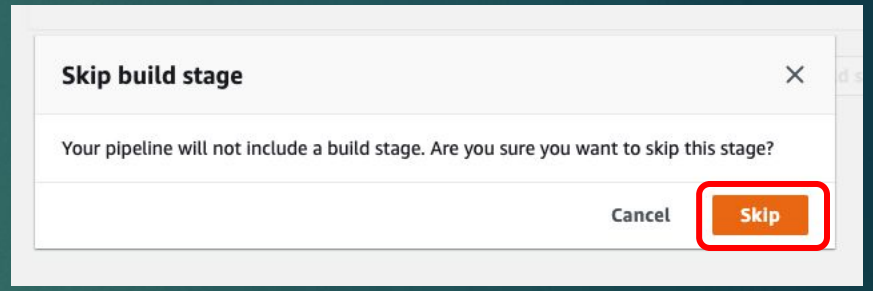


Build Stage

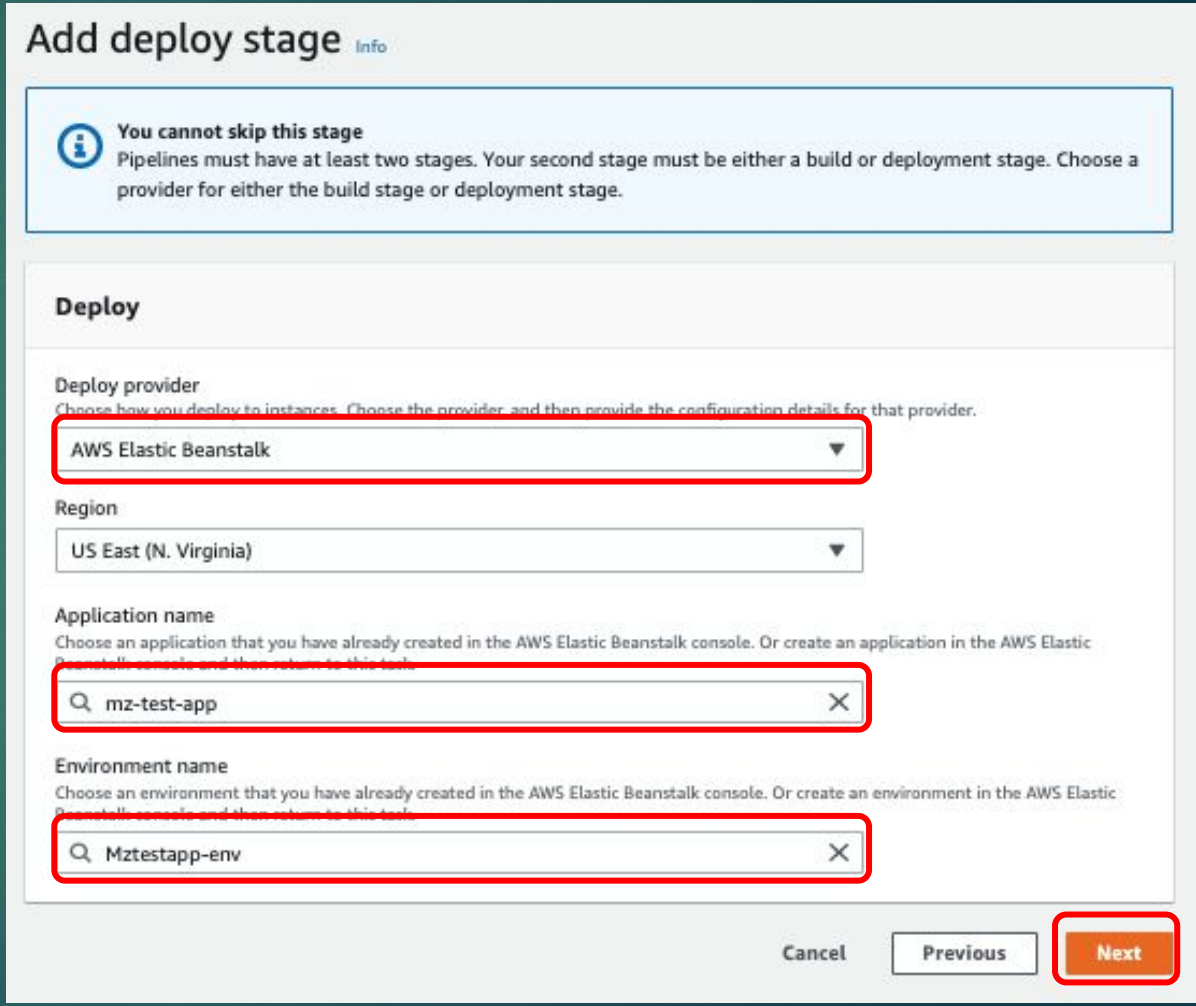
- Click the “Skip build stage” and confirm by clicking “Skip”
- Then click the “Skip test stage”





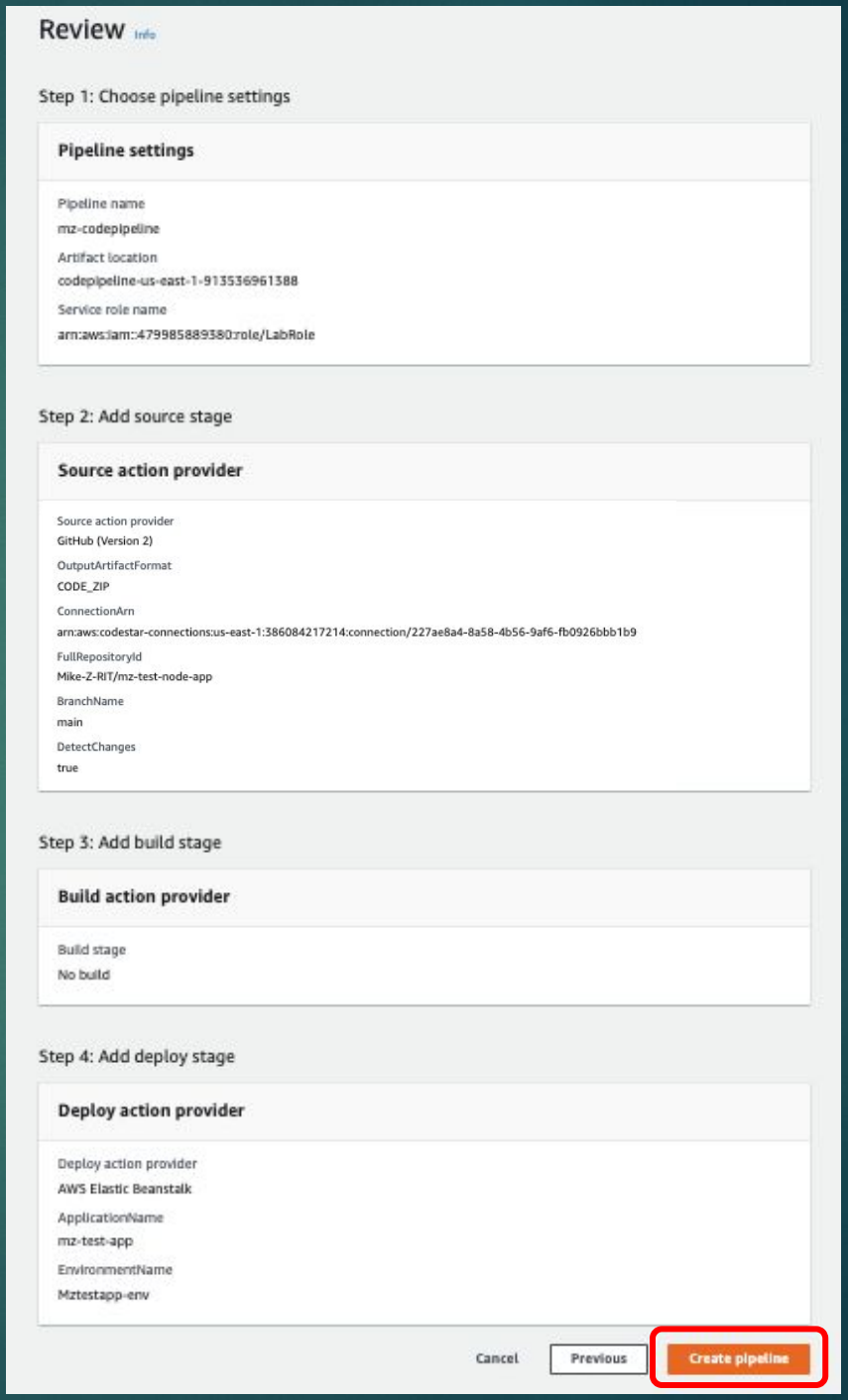
Deploy Stage

- Under "Deploy provider" Select "AWS Elastic Beanstalk"
- Under "Application name" select the application you previously created
- Under "Environment name" select the environment that was previously created
- Click "Next"



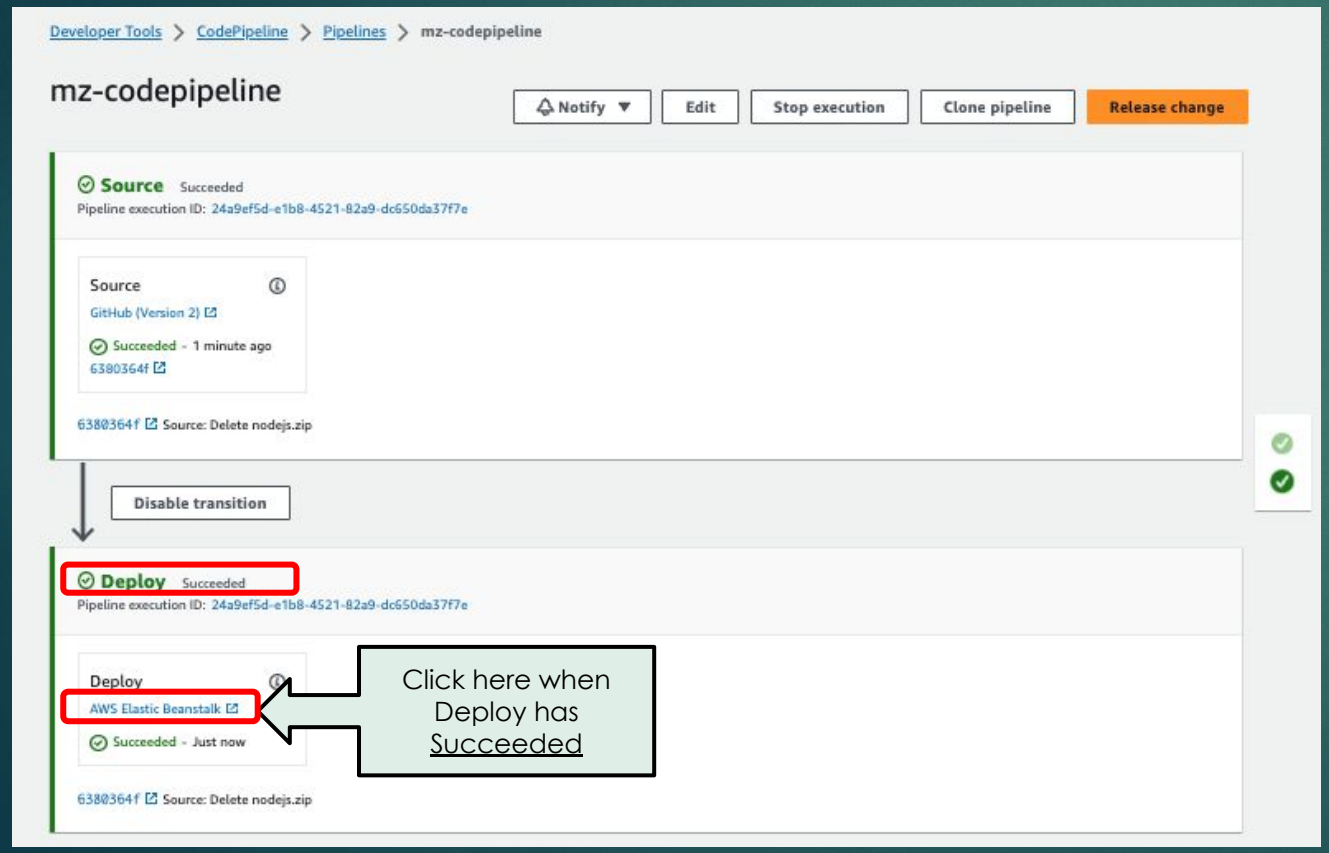
Create your Pipeline

- Click the “Create pipeline”



Build and Deploy

- This will start the Build and Deploy process

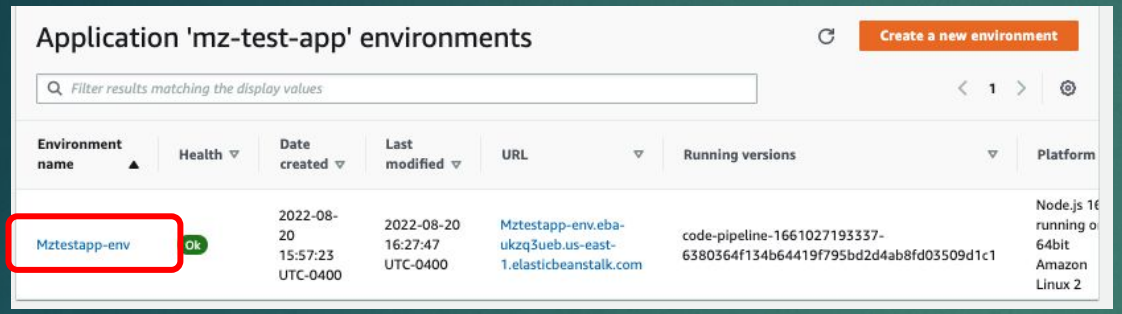


This gets automatically triggered every time you modify the code in Github and commit to the master

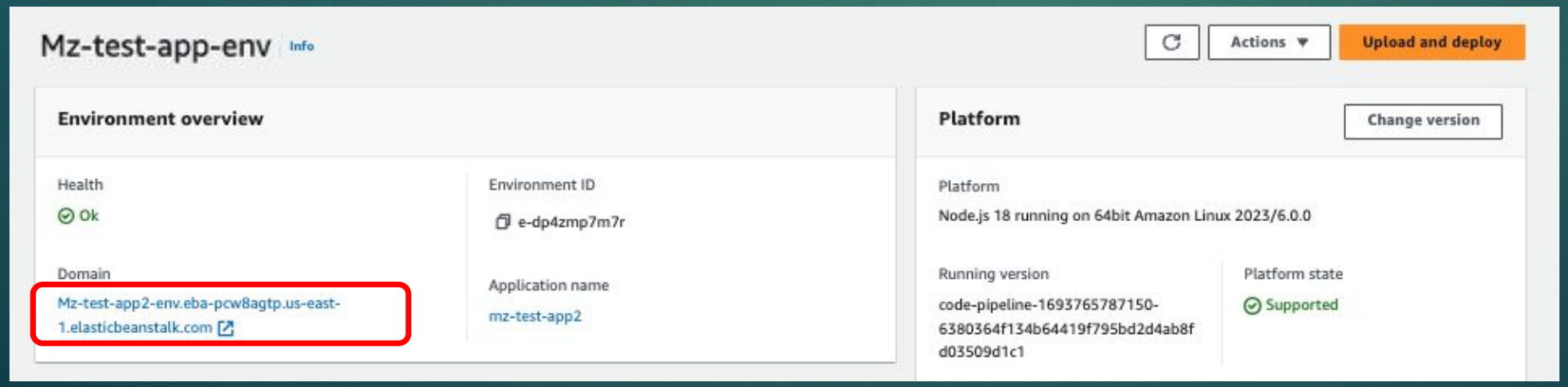
Once the source has been detected, the deploy process will deploy the updated code to your Node application

Create your Pipeline

- Click the link for your web application



- On the Dashboard, click the URL



Test your Web Application

- The sample application (from Github) has been deployed and is running

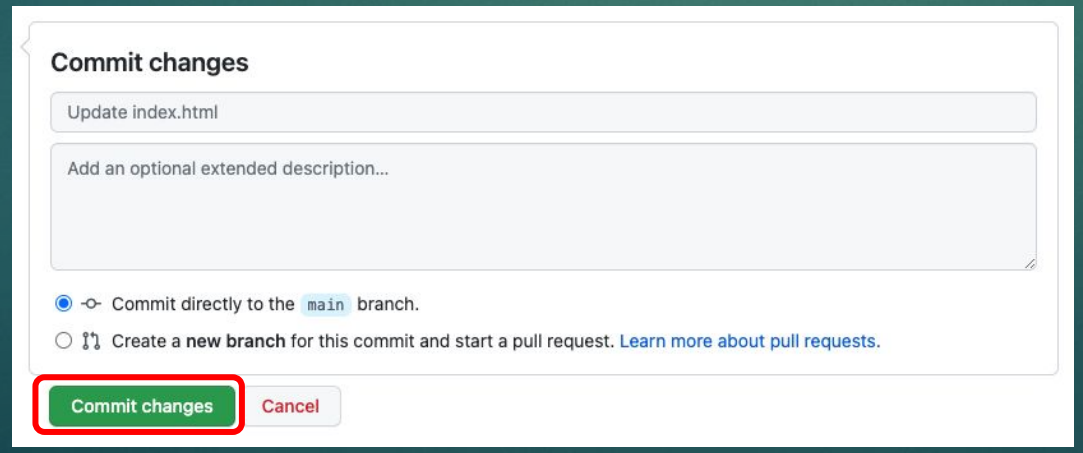
- Congratulations, but you are <u>not</u> done yet
- Now we must make a change to trigger the pipeline to update

Modify your code and commit

- Next, make a change in code
  - You can either do this on your PC or directly in GitHub
- In the index.html, add your RIT ID in the title after "Congratulations"

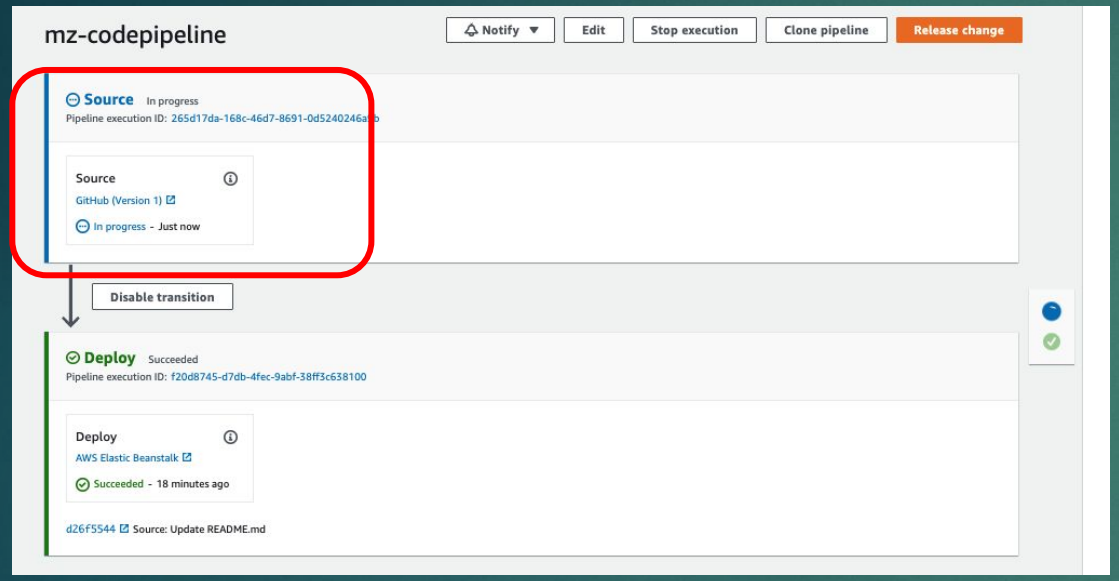
```
76      <div class="textColumn">
77        <h1>Congratulations mszvse</h1>
78        <p>Your first AWS Elastic Beanstalk Node.js application is now running on your own dedicated environment in the AWS Cloud</p>
79        <p>This environment is launched with Elastic Beanstalk Node.js Platform</p>
80      </div>
```

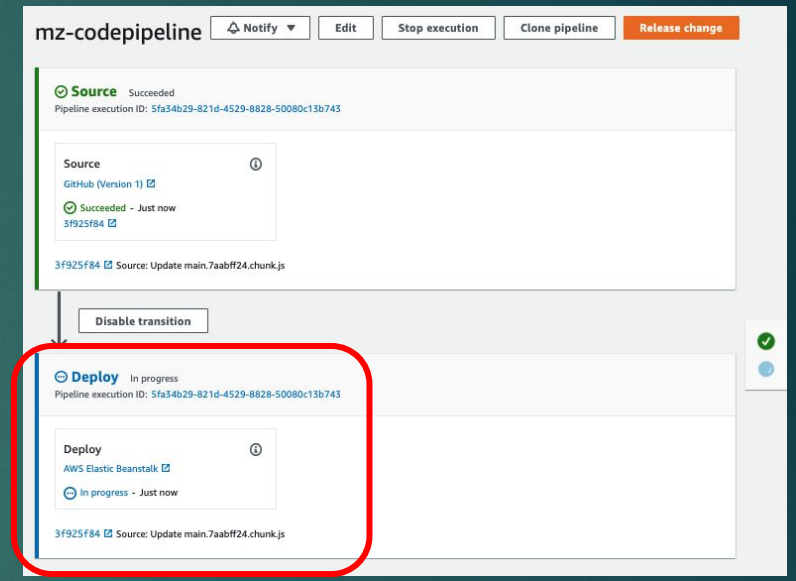
- Either push your change (from your PC) or click "Commit changes" (below)

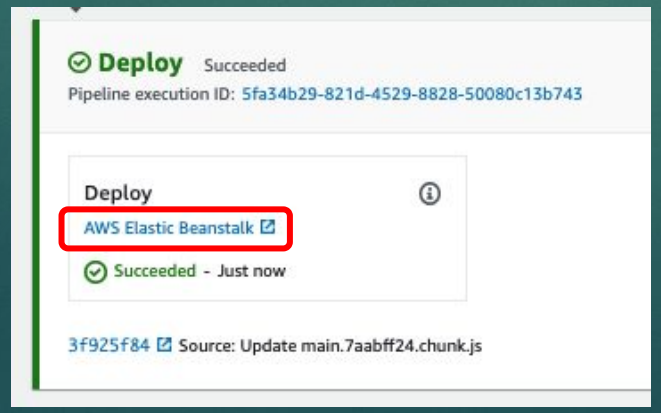


Redeploy your code

- When you push your change, the "Source" will indicate "In Progress" followed by "Deploy" stage







When you see a Succeeded message under Deploy, click the "AWS Elastic Beanstalk" link

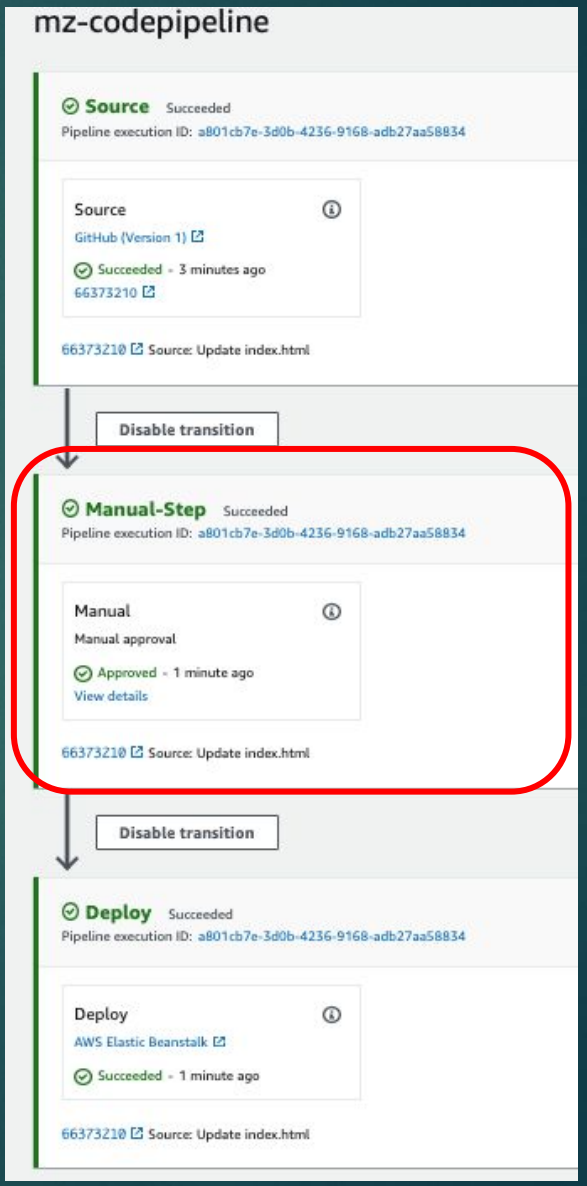
Test your Web Application – Deliverable #1

- You are taken to updated web page of your application
- You now have a fully functioning continuous deployment pipeline!

- Take a screenshot of web page with your update and upload to Assignments > Activity #14 - Create a Continuous Delivery Pipeline

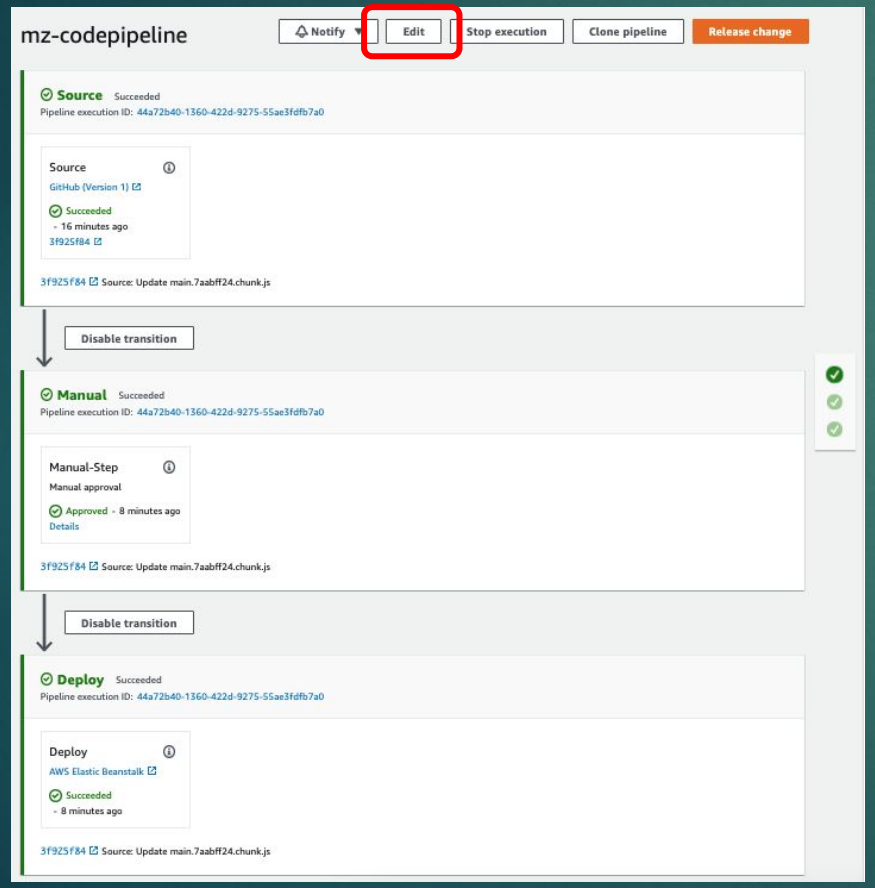
Test your Web Application – Deliverable #2

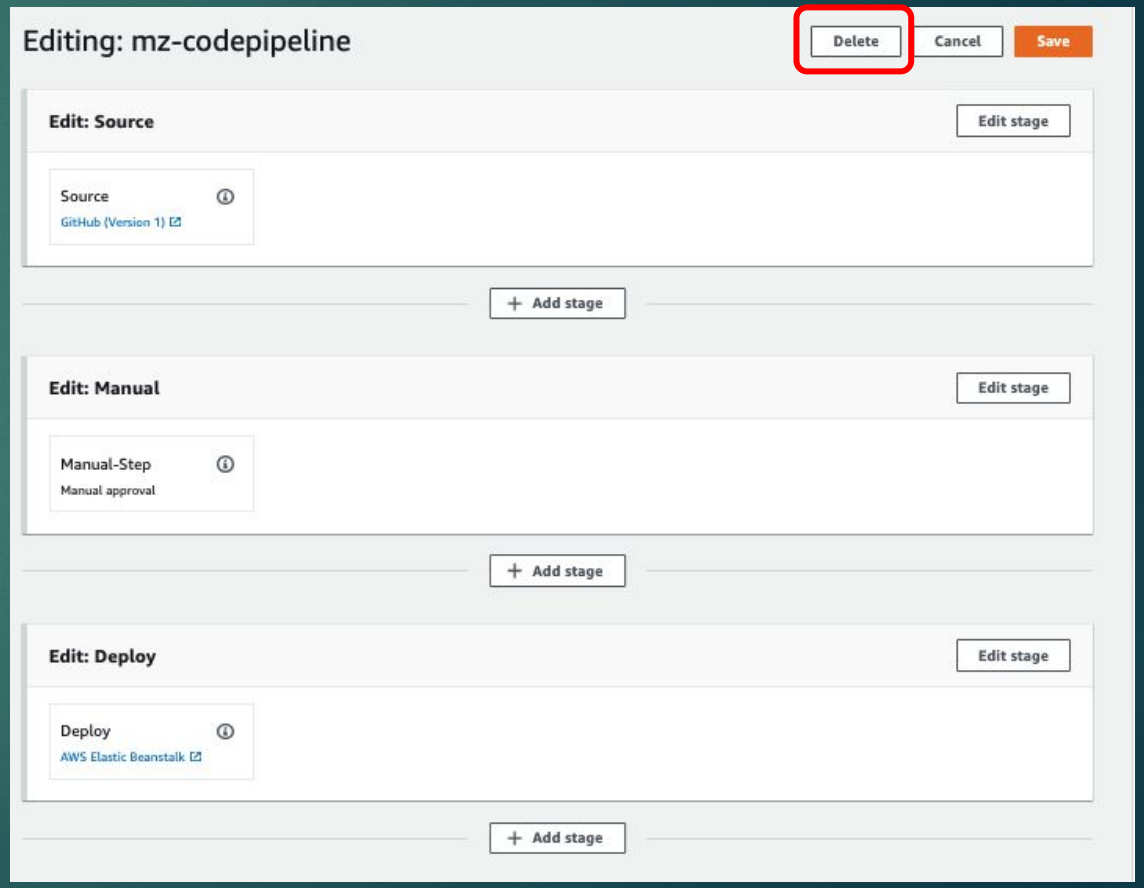
- Next, modify your pipeline to Continuous Delivery
- To do this requires adding a Manual Review Step to your Pipeline
- When you have working, take a screenshot similar to the right
  - Note: You will need to make a code change to re-trigger the pipeline
- Submit screenshot to Assignments > Activity #14 - Create a DevOps Pipeline



Cleanup Pipeline

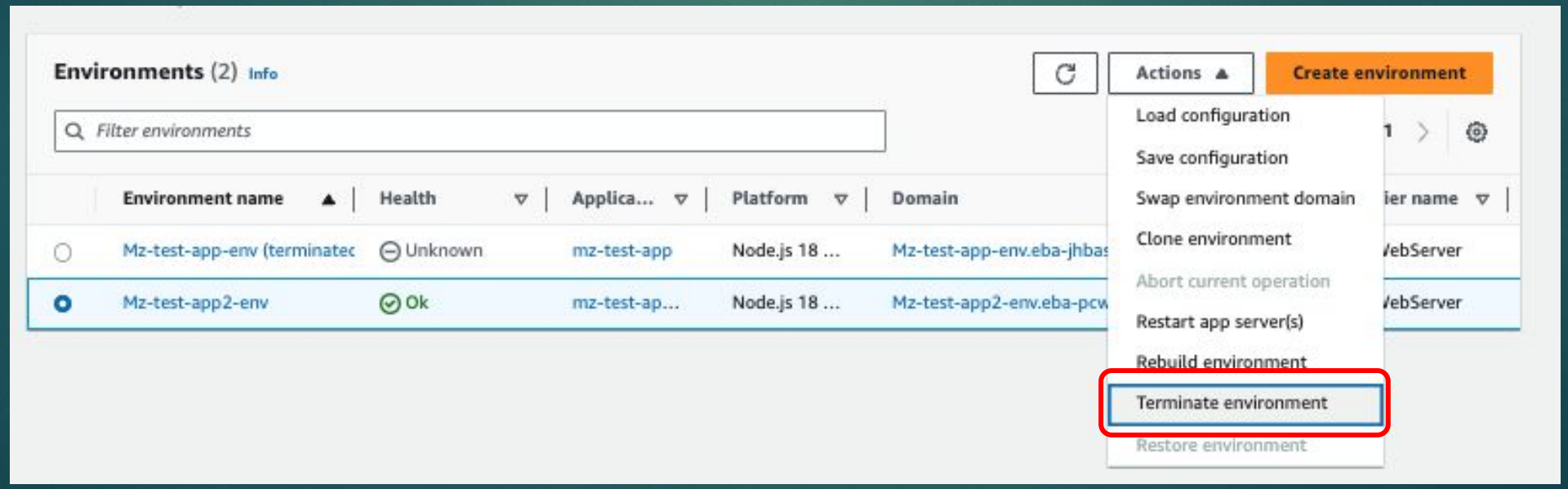
- Go back to CodePipeline and click "Edit"
- Click "Delete" and confirm





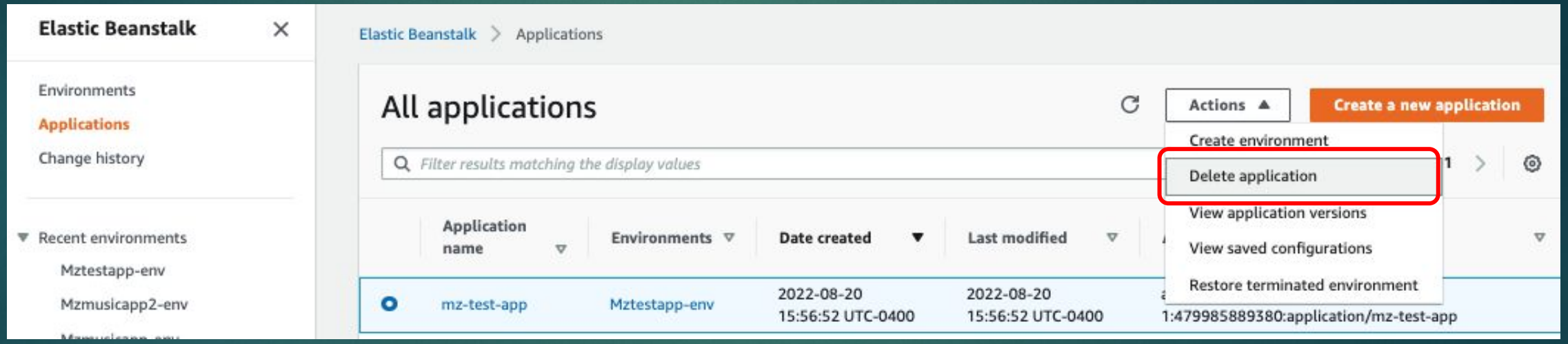
Cleanup Beanstalk

- Go to the Beanstalk console under "Environments" and select your environment name
- Select Actions > Terminate Environment
- Verify by typing the name of the environment and click "Terminate"



Cleanup Beanstalk

- Under "Applications" select your application name
- Select "Actions > Delete application"
- Verify by typing the name of the application and click "Delete"



- You are done!